Templates

Data & Information

- Data Model & Schema (DATA)
- Data Lifecycle & Retention (DLR)
- Data Governance & Lineage (DGL)
- Data Quality & Validation (DQV)
- Search & Indexing (SRCH)
- Caching & Data Access Patterns (CACHE)
- Reporting & Aggregations (RPT)

Data & Information (Definition)

What it is

The Data & Information group defines the system's **canonical data shape, movement, governance, and quality rules**: entities, schemas, validation, retention, lineage, search/index strategy, caching/data-access patterns, and how data is aggregated for reporting. It turns domain concepts into **implementable, enforceable data contracts**.

What it produces

- Canonical **entity + schema definitions** (fields, relationships, constraints)
- **Lifecycle rules** (retention, deletion, archival, legal holds)
- **Governance + lineage** (sources, transformations, access controls, auditability)
- **Validation + quality rules** (schema + semantic checks + DQ monitoring)
- **Search/index design** (what gets indexed, update strategy, query model)
- **Caching/data access patterns** (consistency stance, invalidation, read models)
- **Reporting/aggregation specs** (metrics definitions, rollups, snapshots, access)

What it is used for (exactly)

- Driving DB design and migrations (DATA-04), plus API contracts (API-02)

- Enforcing correctness and compliance (DGP/COMP alignment)
- Preventing schema drift and undocumented transformations
- Making search/discovery deterministic (SRCH templates)
- Making caching safe and consistent (CACHE templates)
- Making reporting accurate and repeatable (RPT templates)

What it is not

- Not UI/UX design (Group 2)
- Not workflow orchestration (Group 3 WFO)
- Not operational deployment docs (Group 8 OPS/BDR)
- Not “truth sources” separate from Canonical Spec; these implement the truth

Scope rules

- Entity names/terms must align with DMG glossary and concept model
- Must specify enforcement points (DB constraints, validation layer, API layer)
- Must separate **canonical data** vs **derived/read models** vs **cached views**
- Must define unknown/optional behavior explicitly (no silent assumptions)

Subcategory meaning (one-line each)

- **DATA**: canonical schema + relationships + constraints + migrations posture
- **DLR**: retention/deletion/archival policies and lifecycle state transitions
- **DGL**: governance + lineage: provenance, transformations, audit, ownership
- **DQV**: validation and quality checks: schema + semantic + monitoring
- **SRCH**: indexing + query model: what’s searchable, update strategy, ranking hooks

- **CACHE**: caching + data access patterns: consistency, invalidation, read models
- **RPT**: reporting & aggregations: rollups, snapshots, metric semantics, access rules

Template Sets

Data & Information — Template Set (Locked v1)

Data Model & Schema (DATA)
- DATA-01 Entity Definitions (canonical fields)
- DATA-02 Relationship Map (ERD text spec)
- DATA-03 Persistence Rules (indexes, constraints)
- DATA-04 Migration Plan (schema evolution)
- DATA-05 Seed Data Spec
- DATA-06 Validation Schemas (Zod/JSON Schema mapping)
- DATA-07 Derived/Read Models Spec (views, projections, denormalized tables)
- DATA-08 Data Access Patterns (transactions, isolation, query boundaries)

Data Lifecycle & Retention (DLR)
- DLR-01 Data Lifecycle States (active/archived/deleted)
- DLR-02 Retention Policy Matrix (by data class/entity)
- DLR-03 Deletion & Erasure Procedures (soft/hard delete)
- DLR-04 Legal Holds & Exceptions Policy
- DLR-05 Archival & Cold Storage Strategy
- DLR-06 Data Minimization Rules (collect/store only what's needed)

Data Governance & Lineage (DGL)
- DGL-01 Data Ownership Map (owner per entity/dataset)
- DGL-02 Lineage Map (sources → transforms → sinks)
- DGL-03 Transformation Rules Catalog (ETL/ELT, normalization)
- DGL-04 Access Controls for Data (who can read/write/export)
- DGL-05 Auditability Requirements (what must be traceable)
- DGL-06 Data Catalog / Dictionary (datasets, meaning, sensitivity)

Data Quality & Validation (DQV)
- DQV-01 Validation Policy (schema vs semantic)
- DQV-02 Data Quality Checks Catalog (rules by entity)
- DQV-03 Input Validation Rules (API/forms → entities)
- DQV-04 Consistency & Integrity Monitoring (drift, anomalies)
- DQV-05 Bad Data Handling (quarantine, repair, backfill)
- DQV-06 DQ Test & Verification Plan

Search & Indexing (SRCH)
- SRCH-01 Search Scope & Surfaces (what is searchable, where)
- SRCH-02 Query Model (filters, ranking signals, facets)
- SRCH-03 Index Update Strategy (sync/async, reindex)
- SRCH-04 Search Result Quality Rules (relevance, freshness, dedupe)
- SRCH-05 Search Abuse Controls (gaming, spam, limits)
- SRCH-06 Search Observability (metrics, logging, evaluation)

Caching & Data Access Patterns (CACHE)

- CACHE-01 Caching Strategy (what to cache, where)
- CACHE-02 Invalidation Rules (events, TTLs, busting)
- CACHE-03 Consistency Model (strong/eventual/stale-while-revalidate)
- CACHE-04 Read/Write Split Rules (read replicas, CQRS if used)
- CACHE-05 Rate/Cost Controls for Reads (hot keys, batching)
- CACHE-06 Cache Failure Behavior (fallbacks, degradation)

Reporting & Aggregations (RPT)

- RPT-01 Reporting Surfaces Inventory (dashboards, exports, admin)
- RPT-02 Metrics Definitions (canonical KPI semantics)
- RPT-03 Aggregation & Rollup Rules (windows, group-bys)
- RPT-04 Snapshotting Strategy (daily/weekly, recompute rules)
- RPT-05 Data Access & Permissions for Reports
- RPT-06 Reporting Data Quality Rules (reconciliation, correctness)

Data Model & Schema (DATA)

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- DATA-01 Entity Definitions (canonical fields)
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- DATA-03 Persistence Rules (indexes, constraints)
- DATA-04 Migration Plan (schema evolution)
- DATA-05 Seed Data Spec
- DATA-06 Validation Schemas (Zod/JSON Schema mapping)
- DATA-07 Derived/Read Models Spec (views, projections, denormalized tables)
- DATA-08 Data Access Patterns (transactions, isolation, query boundaries)

DATA-01

DATA-01 — Entity Definitions (canonical fields)

Header Block

- template_id: DATA-01
- title: Entity Definitions (canonical fields)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-01_Entity_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DMG-01", "DMG-02", "PRD-04", "BRP-01"]
- inputs_required: ["DMG-01", "DMG-02", "PRD-04", "BRP-01", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical data entities and their fields. This is the authoritative entity-level schema reference used by DB modeling, API contracts, validation schemas, and data governance.

Inputs Required

- DMG-01: {{xref:DMG-01}} | OPTIONAL
- DMG-02: {{xref:DMG-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL

- BRP-01: {{xref:BRP-01}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Entity list (minimum 8 for non-trivial products; justify if smaller)
- For each entity:
    - entity_id (stable)
    - name
    - description
    - ownership boundary/service (ARC/SBDT reference)
    - identifiers (primary key, external ids)
    - lifecycle states (if any) pointer (DLR)
    - fields table:
        - field_name
        - type
        - required (true/false)
        - default rule
        - pii_classification (none/low/med/high) or pointer
        - validation rule pointer (DATA-06/DQV)
        - notes
    - invariants (must-always-be-true rules) pointer (DMG-03)

    - audit requirements (if entity is sensitive)

- Coverage check: PRD features reference existing entity_ids where needed

Optional Fields

- Example records | OPTIONAL
- Notes | OPTIONAL

Rules

- Terms and meanings must align to DMG glossary.
- Do not define relationships here (that lives in DATA-02) beyond “foreign key exists” notes.
- PII classification must be explicit for any user-related field.
- Every required field must have either a source or a default rule (no silent required).
- Field names must follow consistent casing convention (tie to standards).

Output Format

1) Entity Index (required)

entity_id	name	owner	primary_key	pii_sensitive	detail_block_present	notes
{{entities[0].id}}	{{entities[0].name}}	{{entities[0].owner}}	{{entities[0].pk}}	{{entities[0].pii_sensitive}}	{{entities[0].block_present}}	{{entities[0].notes}}

2) Entity Detail Blocks (required, one per entity)

{{entities[0].entity_id}} — {{entities[0].name}}

- Description: {{entities[0].description}}

- Owner boundary/service: {{entities[0].owner}}
- Identifiers: {{entities[0].identifiers}}
- Lifecycle pointer: {{xref:DLR-01}} | OPTIONAL
- Invariants pointer: {{xref:DMG-03}} | OPTIONAL

Fields (canonical)

field_name	type	required	default_rule	pii_class	validation_ref	notes
{{entities[0].fields[0].name}}	{{entities[0].fields[0].type}}	{{entities[0].fields[0].required}}	{{entities[0].fields[0].default}}	{{entities[0].fields[0].pii}}	{{entities[0].fields[0].validation_ref}}	{{entities[0].fields[0].notes}}
{{entities[0].fields[1].name}}	{{entities[0].fields[1].type}}	{{entities[0].fields[1].required}}	{{entities[0].fields[1].default}}	{{entities[0].fields[1].pii}}	{{entities[0].fields[1].validation_ref}}	{{entities[0].fields[1].notes}}

Audit Requirements (required if sensitive)

- Must audit create/update/delete: {{entities[0].audit.must_audit}} | OPTIONAL
- Audit fields: {{entities[0].audit.fields}} | OPTIONAL

3) Coverage Checks (required)

- PRD feature → entity coverage complete: {{coverage.features_have_entities}} | OPTIONAL
- Entity IDs unique: {{coverage.unique_entity_ids}}

Cross-References

- Upstream: {{xref:DMG-01}} | OPTIONAL, {{xref:DMG-02}} | OPTIONAL
- Downstream: {{xref:DATA-02}}, {{xref:DATA-03}}, {{xref:DATA-06}} | OPTIONAL, {{xref:DGL-01}} | OPTIONAL, {{xref:API-02}} | OPTIONAL

- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Entity list + core fields + required flags.
- intermediate: Required. Add PII classification and owner mapping.
- advanced: Required. Add invariants pointers and audit requirements.

Unknown Handling

- UNKNOWN_ALLOWED: `example_records`, `notes`, `audit_requirements` (if not sensitive), `lifecycle_states` (if none)
- If any entity lacks primary identifiers → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:
  - required_fields_present == true
  - entities_count >= 8 (or justified)
  - fields_tables_present == true
  - unique_entity_ids == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DATA-02

DATA-02 — Relationship Map (ERD text spec)

Header Block

- template_id: DATA-02
- title: Relationship Map (ERD text spec)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-02_Relationship_Map.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "DMG-02", "DMG-03"]
- inputs_required: ["DATA-01", "DMG-02", "DMG-03", "BRP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical entity relationships (ERD in text form): cardinality, foreign keys, ownership, cascade behaviors, and relationship invariants. This is the authoritative relationship reference for DB schema and API expansions.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DMG-02: {{xref:DMG-02}} | OPTIONAL
- DMG-03: {{xref:DMG-03}} | OPTIONAL

- BRP-01: {{xref:BRP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Relationship list (minimum 12 for non-trivial products; justify if smaller)
- For each relationship:
    - rel_id
    - from_entity_id
    - to_entity_id
    - relationship_type (1:1, 1:N, N:M)
    - fk_field (from → to)
    - requiredness (required/optional)
    - ownership (which entity owns lifecycle)
    - cascade policy (delete/update)
    - referential integrity rule (db enforced vs app enforced)
    - invariants (must hold)
    - notes on query patterns (common joins) | OPTIONAL
- N:M join table specs (if any): join_entity_id + keys

Optional Fields

- Diagram pointer | OPTIONAL
- Notes | OPTIONAL

Rules

- All entity_ids must exist in DATA-01.
- Relationship cardinality must be explicit.
- Cascade policies must align with lifecycle rules (DLR) and privacy deletion constraints (DGP).
- N:M must define join table explicitly (no implicit many-to-many).

Output Format

1) Relationship Registry (canonical)

rel_id	from_entity	to_entity	type	fk_field	required	owner	cascade	enforcement	invariants	notes
rel_01	{{rels[0].from}}	{{rels[0].to}}	{{rels[0].type}}	{{rels[0].fk}}	{{rels[0].required}}	{{rels[0].owner}}	{{rels[0].cascade}}	{{rels[0].enforcement}}	{{rels[0].invariants}}	{{rels[0].notes}}
rel_02	{{rels[1].from}}	{{rels[1].to}}	{{rels[1].type}}	{{rels[1].fk}}	{{rels[1].required}}	{{rels[1].owner}}	{{rels[1].cascade}}	{{rels[1].enforcement}}	{{rels[1].invariants}}	{{rels[1].notes}}

2) N:M Join Tables (required if any)

join_entity_id	left_entity	right_entity	left_fk	right_fk	unique_rule	cascade	notes

{{joins[0].id}}	{{joins[0].left}}	{{joins[0].right}}	{{joins[0].left_fk}}	{{joins[0].right_fk}}	{{joins[0].unique}}	{{joins[0].cascade}}	{{joins[0].notes}}

3) Integrity Constraints Summary (required)

- Must-enforce-in-DB list: {{integrity.db_enforced}}
- App-enforced exceptions (if any): {{integrity.app_enforced}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-01}} | OPTIONAL, {{xref:DMG-03}} | OPTIONAL
- Downstream: {{xref:DATA-03}}, {{xref:DATA-04}} | OPTIONAL, {{xref:API-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Relationship registry with type and fk fields.
- intermediate: Required. Add cascade and enforcement rules.
- advanced: Required. Add invariants and join table specs.

Unknown Handling

- UNKNOWN_ALLOWED: `diagram_pointer`, `notes`, `query_patterns`
- If any relationship references an unknown entity_id → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA

- Pass conditions:
    - required_fields_present == true
    - relationships_count >= 12 (or justified)
    - all_entity_refs_exist == true
    - cardinality_defined == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DATA-03

DATA-03 — Persistence Rules (indexes, constraints)

Header Block

- template_id: DATA-03
- title: Persistence Rules (indexes, constraints)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-03_Persistence_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "DATA-02", "DQV-01"]
- inputs_required: ["DATA-01", "DATA-02", "DQV-01", "SRCH-03", "CACHE-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the persistence-layer rules: indexes, uniqueness constraints, foreign key enforcement, nullability rules, and other DB constraints required to enforce correctness and performance. This makes persistence deterministic and aligned to query/search/caching needs.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-02: {{xref:DATA-02}} | OPTIONAL
- DQV-01: {{xref:DQV-01}} | OPTIONAL

- SRCH-03: {{xref:SRCH-03}} | OPTIONAL
- CACHE-03: {{xref:CACHE-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Constraint catalog (minimum 20 for non-trivial products; justify if smaller)
- Index catalog (minimum 15; justify if smaller)
- For each constraint:
    - con_id
    - entity_id/table
    - type (pk/unique/fk/check/not_null)
    - definition (fields + predicate)
    - enforcement (db/app)
    - rationale (correctness/perf/security)
    - related invariants (DMG-03 pointer) | OPTIONAL
- For each index:
    - idx_id
    - entity_id/table
    - fields
    - type (btree/hash/gin/etc)
    - uniqueness (true/false)
    - query pattern supported

- write impact note (high/med/low)
- maintenance notes | OPTIONAL

Optional Fields

- Partitioning policy | OPTIONAL
- Notes | OPTIONAL

Rules

- Constraints enforce truth; app-only enforcement must be justified.
- Indexes must be tied to real query/search patterns (SRCH/CACHE/API usage).
- Avoid redundant indexes; include write impact note.
- Foreign keys should be DB-enforced unless explicitly incompatible with scale/topology.

Output Format

1) Constraints Catalog (canonical)

con_id	entity	type	definition	enforcement	rationale	invariant_ref	notes
con_01	{{constraints[0].entity}}	{{constraints[0].type}}	{{constraints[0].definition}}	{{constraints[0].enforcement}}	{{constraints[0].rationale}}	{{constraints[0].invariant_ref}}	{{constraints[0].notes}}

con_02	{{constraints[1].entity}}	{{constraints[1].type}}	{{constraints[1].definition}}	{{constraints[1].enforcement}}	{{constraints[1].rationale}}	{{constraints[1].invariant_ref}}	{{constraints[1].notes}}

2) Index Catalog (canonical)

idx_id	entity	fields	index_type	unique	query_pattern	write_impact	notes
idx_01	{{indexes[0].entity}}	{{indexes[0].fields}}	{{indexes[0].type}}	{{indexes[0].unique}}	{{indexes[0].pattern}}	{{indexes[0].impact}}	{{indexes[0].notes}}
idx_02	{{indexes[1].entity}}	{{indexes[1].fields}}	{{indexes[1].type}}	{{indexes[1].unique}}	{{indexes[1].pattern}}	{{indexes[1].impact}}	{{indexes[1].notes}}

3) Nullability & Defaults Rules (required)

- Nullability stance: {{nulls.stance}}
- Defaults stance: {{defaults.stance}}
- Audit timestamps policy: {{defaults.audit_timestamps}} | OPTIONAL

4) Partitioning Policy (optional)

- Applies: {{partitioning.applies}} | OPTIONAL
- Strategy: {{partitioning.strategy}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-01}} | OPTIONAL, {{xref:DATA-02}} | OPTIONAL

- Downstream: {{xref:DATA-04}} | OPTIONAL, {{xref:SRCH-03}} | OPTIONAL, {{xref:CACHE-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Basic constraints + key indexes.
- intermediate: Required. Add rationale and query patterns.
- advanced: Required. Add write impact and partitioning (if applicable).

Unknown Handling

- UNKNOWN_ALLOWED: `partitioning_policy`, `maintenance_notes`, `notes`, `invariant_ref`
- If critical uniqueness constraints are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:
    - required_fields_present == true
    - constraints_count >= 20 (or justified)
    - indexes_count >= 15 (or justified)
    - query_patterns_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DATA-04

DATA-04 — Migration Plan (schema evolution)

Header Block

- template_id: DATA-04
- title: Migration Plan (schema evolution)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-04_Migration_Plan.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "DATA-02", "APIG-06", "REL-04"]
- inputs_required: ["DATA-01", "DATA-02", "DATA-03", "APIG-06", "REL-04", "SBDT-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the deterministic plan for evolving the database schema safely: migration sequencing, expand/contract approach, backfills, verification steps, and rollback posture. This prevents breaking running systems and supports mixed-version deployments.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-02: {{xref:DATA-02}} | OPTIONAL
- DATA-03: {{xref:DATA-03}} | OPTIONAL

- APIG-06: {{xref:APIG-06}} | OPTIONAL
- REL-04: {{xref:REL-04}} | OPTIONAL
- SBDT-06: {{xref:SBDT-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Migration philosophy (expand/contract default stance)
- Migration categories:
    - additive (safe)
    - backfill required
    - destructive (breaking)
- Migration runbook (high level):
    - create migration
    - apply in stage
    - verify
    - apply in prod
- Migration registry (minimum 10 if system evolves; justify if initial)
- For each planned migration:
    - mig_id
    - description
    - affected entities/tables
    - category

    - steps (ordered)
    - backfill required (yes/no)
    - verification queries/checks
    - rollback strategy (rollback/roll-forward)
    - compatibility notes (old+new code)
    - owner
- Safety constraints:
    - no long locks policy
    - batching rules for backfills
    - maintenance window policy (if any)

Optional Fields

- Tooling (Drizzle/Prisma/etc) notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Destructive migrations require explicit approval and a rollback/roll-forward plan.
- Backfills must be resumable and observable.
- All migrations must include verification steps.
- Migration steps must align with deployment constraints (SBDT-06).

Output Format

1) Migration Philosophy (required)

- Default stance: {{philosophy.default}} (expand/contract)
- Compatibility rule: {{philosophy.compatibility}}
- Verification requirement: {{philosophy.verification}}

2) Migration Runbook (required)

1. Create migration: {{runbook.create}}
2. Stage apply: {{runbook.stage_apply}}
3. Verify: {{runbook.verify}}
4. Prod apply: {{runbook.prod_apply}}
5. Post-apply monitor: {{runbook.monitor}} | OPTIONAL

3) Migration Registry (canonical)

mig_id	description	tables	category	steps	backfill	verification	rollback	compat_notes	owner	notes
mig_001	{{migs[0].desc}}	{{migs[0].tables}}	{{migs[0].category}}	{{migs[0].steps}}	{{migs[0].backfill}}	{{migs[0].verify}}	{{migs[0].rollback}}	{{migs[0].compat}}	{{migs[0].owner}}	{{migs[0].notes}}
mig_002	{{migs[1].desc}}	{{migs[1].tables}}	{{migs[1].category}}	{{migs[1].steps}}	{{migs[1].backfill}}	{{migs[1].verify}}	{{migs[1].rollback}}	{{migs[1].compat}}	{{migs[1].owner}}	{{migs[1].notes}}

4) Safety Constraints (required)

- No long locks policy: {{safety.no_long_locks}}

- Backfill batching: {{safety.backfill_batching}}
- Maintenance window policy: {{safety.maintenance_window}} | OPTIONAL

Cross-References

- Upstream: {{xref:APIG-06}} | OPTIONAL, {{xref:SBDT-06}} | OPTIONAL
- Downstream: {{xref:REL-04}} | OPTIONAL, {{xref:BDR-03}} | OPTIONAL, {{xref:QA-05}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Philosophy + runbook + registry skeleton.
- intermediate: Required. Add verification and compatibility notes per migration.
- advanced: Required. Add rollback/roll-forward posture and safety constraints.

Unknown Handling

- UNKNOWN_ALLOWED: `tooling_notes`, `maintenance_window`, `notes`
- If any migration lacks verification steps → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:
    - required_fields_present == true

- runbook_present == true
- registry_present == true
- verification_present_for_all_migrations == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DATA-05

DATA-05 — Seed Data Spec

Header Block

- template_id: DATA-05
- title: Seed Data Spec
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-05_Seed_Data_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "PMAD-01", "DSYS-01"]
- inputs_required: ["DATA-01", "PMAD-01", "DSYS-01", "ENV-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic seed data required for the system to run in dev/stage (and sometimes prod): default roles, permissions, system settings, reference tables, feature flags defaults, and any canonical lookup sets.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- PMAD-01: {{xref:PMAD-01}} | OPTIONAL
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- ENV-01: {{xref:ENV-01}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Seed set inventory (minimum 8 seed sets; justify if smaller)
- For each seed set:
    - seed_id
    - target entity/table
    - environment applicability (dev/stage/prod)
    - records definition (what values, keyed by stable IDs)
    - idempotency rule (safe to rerun)
    - dependencies (must exist before)
    - owner
    - verification query (how to confirm seeded)
- Secrets exclusion rule (never seed secrets)
- Environment safety rules (prod seeding approval policy)

Optional Fields

- Data generator rules (synthetic data) | OPTIONAL
- Notes | OPTIONAL

Rules

- Seed data must be idempotent and safe to rerun.

- Seed IDs must be stable; don't rely on auto-increment values.
- Never seed secrets; reference ENV secrets management.
- Production seeding must be explicit and approval-gated.

Output Format

1) Seed Sets (canonical)

seed_id	table/entity	envs	records_keying	dependencies	idempotent_rule	verify_query	owner	notes
seed_roles	{{seeds[0].table}}	{{seeds[0].envs}}	{{seeds[0].keying}}	{{seeds[0].deps}}	{{seeds[0].idem}}	{{seeds[0].verify}}	{{seeds[0].owner}}	{{seeds[0].notes}}
seed_settings	{{seeds[1].table}}	{{seeds[1].envs}}	{{seeds[1].keying}}	{{seeds[1].deps}}	{{seeds[1].idem}}	{{seeds[1].verify}}	{{seeds[1].owner}}	{{seeds[1].notes}}

2) Seed Record Blocks (required, one per seed set)

{{seeds[0].seed_id}} — {{seeds[0].table}}

- Environments: {{seeds[0].envs}}
- Dependencies: {{seeds[0].deps}}
- Idempotency: {{seeds[0].idem}}
- Verification query: {{seeds[0].verify}}

Records (canonical, keyed)

```
records:
 {{seeds[0].records[0].key}}:
   {{seeds[0].records[0].field1}}: {{seeds[0].records[0].value1}}
   {{seeds[0].records[0].field2}}: {{seeds[0].records[0].value2}}
```

3) Safety Rules (required)

- Never seed secrets: {{safety.no_secrets}}
- Prod seeding policy: {{safety.prod_policy}}
- Rollback policy (if prod seeded): {{safety.rollback}} | OPTIONAL

Cross-References

- Upstream: {{xref:PMAD-01}} | OPTIONAL, {{xref:ENV-01}} | OPTIONAL
- Downstream: {{xref:IMP-01}} | OPTIONAL, {{xref:DX-01}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
- Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Seed sets table + at least 1 record block example.
- intermediate: Required. Add verification queries and dependency ordering.
- advanced: Required. Add prod policy and rollback posture.

Unknown Handling

- UNKNOWN_ALLOWED: `data_generator_rules`, `notes`, `rollback_policy`
- If any seed set lacks idempotency rule → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:

- required_fields_present == true
- seed_sets_present == true
- record_blocks_present == true
- idempotency_defined_for_all == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DATA-06

DATA-06 — Validation Schemas (Zod/JSON Schema mapping)

Header Block

- template_id: DATA-06
- title: Validation Schemas (Zod/JSON Schema mapping)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-06_Validation_Schemas.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "DQV-01", "API-02"]
- inputs_required: ["DATA-01", "DQV-01", "API-02", "FORM-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the validation schema layer mapping from canonical entity fields to enforceable schemas (Zod/JSON Schema/other): what is validated where, which schemas exist, and how schema versions track entity evolution.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DQV-01: {{xref:DQV-01}} | OPTIONAL
- API-02: {{xref:API-02}} | OPTIONAL

- FORM-01: {{xref:FORM-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Schema registry (minimum 12 schemas for non-trivial products)
- For each schema:
    - schema_id
    - kind (entity_create/entity_update/api_request/api_response/form_input)
    - target entity_id or endpoint_id
    - schema language (zod/jsonschema)
    - version
    - required fields
    - optional fields
    - validation rules (field-level + cross-field)
    - error mapping policy (reason_code for failures)
    - reuse policy (shared vs per-endpoint)
    - enforcement point (client/server/both)
- Versioning rule (schema changes align to DATA-04/APIG-06)
- Coverage check: all API writes and all forms have schemas

Optional Fields

- Codegen notes | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Schema enforcement must be explicit; “validated somewhere” is not allowed.
- Cross-field semantic validation must be captured (DQV).
- Validation errors must map to error model (ERR) and accessible error messaging (A11Y/CDX pointers).

Output Format

1) Applicability

- applies: {{schemas.applies}} (true/false)
- 00_NA (if not applies): {{schemas.na_block}} | OPTIONAL

2) Schema Registry (canonical)

schema_id	kind	target	language	version	enforced_at	error_mapping	notes
sch_01	{{registry[0].kind}}	{{registry[0].target}}	{{registry[0].lang}}	{{registry[0].version}}	{{registry[0].enforced_at}}	{{registry[0].error_map}}	{{registry[0].notes}}
sch_02	{{registry[1].kind}}	{{registry[1].target}}	{{registry[1].lang}}	{{registry[1].version}}	{{registry[1].enforced_at}}	{{registry[1].error_map}}	{{registry[1].notes}}

3) Schema Blocks (required, one per schema_id)

{{registry[0].schema_id}}

- Kind: {{registry[0].kind}}
- Target: {{registry[0].target}}
- Version: {{registry[0].version}}
- Enforced at: {{registry[0].enforced_at}}
- Required fields: {{schema_detail[registry[0].schema_id].required_fields}}
- Optional fields: {{schema_detail[registry[0].schema_id].optional_fields}}
- Cross-field rules: {{schema_detail[registry[0].schema_id].cross_field_rules}} | OPTIONAL
- Error mapping: {{schema_detail[registry[0].schema_id].error_mapping}}
- Reuse policy: {{schema_detail[registry[0].schema_id].reuse}}

4) Versioning Rules (required if applies)

- Align to DATA-04: {{versioning.data_alignment}}
- Align to APIG-06: {{versioning.api_alignment}} | OPTIONAL

5) Coverage Checks (required if applies)

- All API writes have schemas: {{coverage.api_writes}}
- All forms have schemas: {{coverage.forms}}

Cross-References

- Upstream: {{xref:DQV-01}} | OPTIONAL, {{xref:API-02}} | OPTIONAL
- Downstream: {{xref:DQV-03}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Registry + coverage checks.
- advanced: Required if applies. Add versioning alignment and semantic rules mapping.

Unknown Handling

- UNKNOWN_ALLOWED: `codegen_notes`, `notes`, `cross_field_rules` (if none), `reuse_policy` (if per-target)
- If applies == true and coverage checks are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_registry_present == true
    - if_applies_then_coverage_present == true
    - versioning_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DATA-07

DATA-07 — Derived/Read Models Spec (views, projections, denormalized tables)

Header Block

- template_id: DATA-07
- title: Derived/Read Models Spec (views, projections, denormalized tables)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-07_Derived_Read_Models_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-01", "DATA-02", "SRCH-03", "CACHE-03"]
- inputs_required: ["DATA-01", "DATA-02", "SRCH-03", "CACHE-03", "WFO-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the derived/read models used to optimize reads and UX: DB views, materialized views, projections, denormalized tables, and read-optimized DTOs. This makes derived data explicit, controlled, and consistent with canonical entities.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-02: {{xref:DATA-02}} | OPTIONAL
- SRCH-03: {{xref:SRCH-03}} | OPTIONAL

- CACHE-03: {{xref:CACHE-03}} | OPTIONAL
- WFO-01: {{xref:WFO-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Read model registry (minimum 6 if system has non-trivial reads)
- For each read model:
    - read_model_id
    - type (view/materialized_view/projection/denorm_table/dto)
    - purpose / query pattern served
    - source entities (canonical)
    - fields included (list)
    - refresh/update strategy:
        - sync on write
        - async projection job
        - scheduled recompute
    - freshness expectation (seconds/minutes/hours)
    - consistency posture (strong/eventual)
    - invalidation triggers (events/updates)
    - ownership (service/boundary)
    - access controls (who can read)

    - failure behavior (fallback to canonical query? stale allowed?)
    - verification query/check (how to validate correctness)
- Coverage check: key read endpoints/screens reference a defined read model or explicitly “canonical only”

Optional Fields

- Storage/cost notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Derived data must never become a hidden truth source; canonical entities remain authoritative.
- Update strategy must be explicit; “kept in sync” is not acceptable without mechanism.
- Access control must match PMAD/DGP policies; do not broaden access via denorm.
- Freshness and consistency must be declared and aligned with UX expectations.

Output Format

1) Applicability

- applies: {{read_models.applies}} (true/false)
- 00_NA (if not applies): {{read_models.na_block}} | OPTIONAL

2) Read Model Registry (canonical)

read_mo	type	purpose	source_entities	update_strategy	freshness	consistency	owner	access	fallback	verify	notes

del _id											
rm _01	{{mo dels[0].ty pe}}	{{mod els[0]. purpo se}}	{{mod els[0]. sourc es}}	{{mo dels[0].up date} }	{{mod els[0].f reshn ess}}	{{mode ls[0].co nsisten cy}}	{{mo dels[0].ow ner}}	{{mo dels[0].ac cess} }	{{mod els[0]. fallba ck}}	{{mo dels[0].ver ify}}	{{mo dels[0].no tes}}
rm _02	{{mo dels[1].ty pe}}	{{mod els[1]. purpo se}}	{{mod els[1]. sourc es}}	{{mo dels[1].up date} }	{{mod els[1].f reshn ess}}	{{mode ls[1].co nsisten cy}}	{{mo dels[1].ow ner}}	{{mo dels[1].ac cess} }	{{mod els[1]. fallba ck}}	{{mo dels[1].ver ify}}	{{mo dels[1].no tes}}

3) Read Model Detail Block (required, one per read_model_id)

{{models[0].read_model_id}} — {{models[0].purpose}}

- Type: {{models[0].type}}
- Query pattern served: {{models[0].purpose}}
- Source entities: {{models[0].sources}}
- Fields included: {{models[0].fields}}
- Update strategy: {{models[0].update}}
- Freshness expectation: {{models[0].freshness}}
- Consistency posture: {{models[0].consistency}}
- Invalidation triggers: {{models[0].invalidate}}
- Access controls: {{models[0].access}}
- Failure behavior: {{models[0].failure_behavior}}
- Verification: {{models[0].verify}}

4) Coverage Checks (required if applies)

- Key read surfaces mapped: {{coverage.read_surfaces_mapped}}
- All models have update strategy: {{coverage.update_defined}}
- Access controls defined: {{coverage.access_defined}}

Cross-References

- Upstream: {{xref:DATA-01}} | OPTIONAL, {{xref:SRCH-03}} | OPTIONAL, {{xref:CACHE-03}} | OPTIONAL
- Downstream: {{xref:CACHE-01}} | OPTIONAL, {{xref:RPT-03}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Registry + update strategy + freshness.
- advanced: Required if applies. Add access controls, fallback behavior, and verification checks.

Unknown Handling

- UNKNOWN_ALLOWED: `storage_cost_notes`, `notes`
- If applies == true and any model lacks update strategy or access controls → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA

- Pass conditions:
    - required_fields_present == true
    - if_applies_then_models_present == true
    - update_defined == true
    - access_defined == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DATA-08

DATA-08 — Data Access Patterns (transactions, isolation, query boundaries)

Header Block

- template_id: DATA-08
- title: Data Access Patterns (transactions, isolation, query boundaries)
- type: data_model_schema
- template_version: 1.0.0
- output_path: 10_app/data/DATA-08_Data_Access_Patterns.md
- compliance_gate_id: TMP-05.PRIMARY.DATA
- upstream_dependencies: ["DATA-03", "BPLAT-04", "CACHE-03"]
- inputs_required: ["DATA-03", "BPLAT-04", "CACHE-03", "WFO-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the allowed data access patterns: transaction boundaries, isolation levels, query shape rules, read/write separation, and how services should access data consistently and safely (especially under concurrency and caching).

Inputs Required

- DATA-03: {{xref:DATA-03}} | OPTIONAL
- BPLAT-04: {{xref:BPLAT-04}} | OPTIONAL
- CACHE-03: {{xref:CACHE-03}} | OPTIONAL

- WFO-03: {{xref:WFO-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Transaction policy:
    - when to open transactions
    - max duration rule
    - nesting policy
- Isolation policy:
    - default isolation level
    - when to elevate isolation
    - phantom/read anomalies stance
- Query boundary rules:
    - service owns its DB access (no cross-service DB access)
    - allowed join depth
    - N+1 avoidance policy
- Write patterns:
    - insert/update/delete patterns
    - soft delete rules pointer (DLR)
    - audit stamping policy
- Read patterns:
    - use of read models (DATA-07)

    - caching interactions (CACHE)
- Error handling and retries alignment (ERR-05)
- Verification checklist

Optional Fields

- ORM-specific conventions | OPTIONAL
- Notes | OPTIONAL

Rules

- Transaction boundaries must align with idempotency/concurrency rules (WFO-03).
- No cross-service DB reads unless explicitly approved and documented.
- Retry within transactions must be careful; define safe retry policy.
- Isolation elevation must be justified and limited.

Output Format

1) Transaction Policy (required)

- When to use transactions: {{tx.when}}
- Max duration: {{tx.max_duration}}
- Nesting policy: {{tx.nesting}} | OPTIONAL

2) Isolation Policy (required)

- Default level: {{isolation.default}}
- Elevated level cases: {{isolation.elevate_cases}}

- Anomaly stance: {{isolation.anomalies}} | OPTIONAL

3) Query Boundary Rules (required)

- Ownership rule: {{query.ownership}}
- Join depth rule: {{query.join_depth}}
- N+1 avoidance: {{query.n_plus_one}}

4) Write Patterns (required)

- Write method patterns: {{writes.patterns}}
- Soft delete pointer: {{xref:DLR-03}} | OPTIONAL
- Audit stamping: {{writes.audit_stamping}} | OPTIONAL

5) Read Patterns (required)

- Read model usage: {{reads.read_models}}
- Cache interaction pointer: {{xref:CACHE-03}} | OPTIONAL
- Read replica usage (if any): {{reads.replicas}} | OPTIONAL

6) Error/Retry Alignment (required)

- Retry policy pointer: {{xref:ERR-05}} | OPTIONAL
- Safe retry rule in DB operations: {{retries.safe_rule}}

7) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}

- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:WFO-03}} | OPTIONAL, {{xref:CACHE-03}} | OPTIONAL
- Downstream: {{xref:API-02}} | OPTIONAL, {{xref:BPLAT-01}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Transaction + query boundary basics.
- intermediate: Required. Add isolation and read/write patterns.
- advanced: Required. Add retry alignment and verification checklist.

Unknown Handling

- UNKNOWN_ALLOWED: `orm_conventions`, `notes`, `read_replica_usage`
- If default isolation or ownership rule is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DATA
- Pass conditions:
    - required_fields_present == true
    - transaction_policy_present == true

- isolation_policy_present == true
- query_boundary_rules_present == true
- retry_alignment_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

Data Lifecycle & Retention (DLR)

Data Lifecycle & Retention (DLR)

- DLR-01 Data Lifecycle States (active/archived/deleted)
- DLR-02 Retention Policy Matrix (by data class/entity)
- DLR-03 Deletion & Erasure Procedures (soft/hard delete)
- DLR-04 Legal Holds & Exceptions Policy
- DLR-05 Archival & Cold Storage Strategy
- DLR-06 Data Minimization Rules (collect/store only what’s needed)

DLR-01

DLR-01 — Data Lifecycle States (active/archived/deleted)

Header Block

- template_id: DLR-01
- title: Data Lifecycle States (active/archived/deleted)
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-01_Data_Lifecycle_States.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["DATA-01", "DGP-02", "BRP-01"]
- inputs_required: ["DATA-01", "DGP-02", "BRP-01", "RISK-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical lifecycle states for persisted data (e.g., active → archived → deleted), what each state means, and what system behaviors are allowed in each state. This enables consistent soft-delete, archival, and retention enforcement.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DGP-02: {{xref:DGP-02}} | OPTIONAL
- BRP-01: {{xref:BRP-01}} | OPTIONAL

- RISK-02: {{xref:RISK-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Lifecycle state set (minimum: active, archived, deleted)
- State definitions (meaning and constraints)
- State transition rules:
    - allowed transitions
    - disallowed transitions
    - who/what triggers transitions
- Per-entity applicability:
    - which entities have lifecycle states
    - what field represents lifecycle (status, deleted_at, etc.)
- Behavioral constraints per state:
    - readable? writable? searchable? reportable?
- Verification checklist

Optional Fields

- Additional states (pending, suspended) | OPTIONAL
- Notes | OPTIONAL

Rules

- “Deleted” must define whether it is soft-delete vs hard-delete (pointer to DLR-03).
- Archived data access must be explicitly defined (read-only vs hidden).
- Search and reporting must respect lifecycle constraints.
- Transitions must be auditable for sensitive entities.

Output Format

1) States (required)

state	meaning	allowed_access	constraints	notes
active	{{states.active.meaning}}	{{states.active.access}}	{{states.active.constraints}}	{{states.active.notes}}
archived	{{states.archived.meaning}}	{{states.archived.access}}	{{states.archived.constraints}}	{{states.archived.notes}}
deleted	{{states.deleted.meaning}}	{{states.deleted.access}}	{{states.deleted.constraints}}	{{states.deleted.notes}}

2) Transition Rules (required)

from	to	allowed	trigger	audit_required	notes
active	archived	true	{{transitions.active_to_archived.trigger}}	{{transitions.active_to_archived.audit}}	{{transitions.active_to_archived.notes}}
archived	deleted	true	{{transitions.archived_to_deleted.trigger}}	{{transitions.archived_to_deleted.audit}}	{{transitions.archived_to_deleted.notes}}
deleted	active	false	n/a	true	not allowed

3) Entity Applicability (required)

entity_id	has_lifecycle	lifecycle_field	allowed_states	notes

{{entities[0].id}}	{{entities[0].has_lifecycle}}	{{entities[0].field}}	{{entities[0].states}}	{{entities[0].notes}}

4) Behavioral Constraints (required)

- Read rules by state: {{behavior.read_rules}}
- Write rules by state: {{behavior.write_rules}}
- Search rules by state: {{behavior.search_rules}}
- Reporting rules by state: {{behavior.reporting_rules}} | OPTIONAL

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-01}} | OPTIONAL, {{xref:DGP-02}} | OPTIONAL
- Downstream: {{xref:DLR-02}}, {{xref:DLR-03}}, {{xref:SRCH-01}} | OPTIONAL, {{xref:RPT-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. State definitions + basic transitions.
- intermediate: Required. Add entity applicability and behavioral constraints.
- advanced: Required. Add audit requirements and lifecycle enforcement points.

Unknown Handling

- UNKNOWN_ALLOWED: `additional_states`, `notes`
- If entity applicability is UNKNOWN for lifecycle-managed entities → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
  - required_fields_present == true
  - states_present == true
  - transitions_present == true
  - entity_applicability_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DLR-02

DLR-02 — Retention Policy Matrix (by data class/entity)

Header Block

- template_id: DLR-02
- title: Retention Policy Matrix (by data class/entity)
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-02_Retention_Policy_Matrix.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["DGP-01", "DATA-01", "DLR-01"]
- inputs_required: ["DGP-01", "DATA-01", "DLR-01", "COMP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the canonical retention periods and deletion requirements for each entity/data class: how long data is kept, what triggers retention timers, what rules apply for archival, and what legal/compliance constraints exist.

Inputs Required

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DLR-01: {{xref:DLR-01}} | OPTIONAL

- COMP-01: {{xref:COMP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Retention matrix entries (minimum 15 for non-trivial products)
- For each entry:
    - entity_id / dataset
    - data_class (PII level / regulated type)
    - retention_period (duration)
    - retention_start_trigger (created_at, last_active_at, closed_at, etc.)
    - storage tier (hot/cold) pointer (DLR-05/STORE)
    - deletion type (soft/hard/anonymize)
    - legal hold override (yes/no)
    - access constraints while retained
    - export requirements (if any)
    - owner
- Compliance notes (regulatory requirements)
- Verification checklist

Optional Fields

- Region-specific retention differences | OPTIONAL

- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Retention must reference data classification (DGP/DGL).
- Legal holds override deletion and must be explicitly defined.
- Deletion type must align with DLR-03 procedures.
- Reporting and search retention must align to lifecycle state constraints.

Output Format

1) Applicability

- applies: {{retention.applies}} (true/false)
- 00_NA (if not applies): {{retention.na_block}} | OPTIONAL

2) Retention Matrix (canonical)

entity_id	data_class	retention_period	start_trigger	storage_tier	deletion_type	legal_hold_override	access_constraints	export_req	owner	notes
{{matrix[0].entity}}	{{matrix[0].class}}	{{matrix[0].period}}	{{matrix[0].trigger}}	{{matrix[0].tier}}	{{matrix[0].deletion}}	{{matrix[0].legal_hold}}	{{matrix[0].access}}	{{matrix[0].export}}	{{matrix[0].owner}}	{{matrix[0].notes}}
{{matrix[1].entity}}	{{matrix[1].class}}	{{matrix[1].period}}	{{matrix[1].trigger}}	{{matrix[1].tier}}	{{matrix[1].deletion}}	{{matrix[1].legal_hold}}	{{matrix[1].access}}	{{matrix[1].export}}	{{matrix[1].owner}}	{{matrix[1].notes}}

3) Compliance Notes (required if applies)

- Regulated data constraints: {{compliance.regulated}} | OPTIONAL
- Default minimum retention: {{compliance.default_min}} | OPTIONAL
- Evidence required: {{compliance.evidence}} | OPTIONAL

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DGP-01}} | OPTIONAL, {{xref:DLR-01}} | OPTIONAL
- Downstream: {{xref:DLR-03}}, {{xref:DLR-04}}, {{xref:DLR-05}} | OPTIONAL, {{xref:STORE-*}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Matrix + start triggers + deletion type.
- advanced: Required if applies. Add legal holds and compliance evidence notes.

Unknown Handling

- UNKNOWN_ALLOWED: `region_differences`, `notes`, `export_requirements`, `compliance_notes`

- If applies == true and any entity lacks retention_period → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_matrix_present == true
    - start_triggers_present == true
    - deletion_types_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DLR-03

DLR-03 — Deletion & Erasure Procedures (soft/hard delete)

Header Block

- template_id: DLR-03
- title: Deletion & Erasure Procedures (soft/hard delete)
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-03_Deletion_Erasure_Procedures.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["DLR-01", "DLR-02", "DGP-02", "ERR-05"]
- inputs_required: ["DLR-01", "DLR-02", "DGP-02", "DGP-04", "DATA-02", "ERR-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the deterministic procedures for data deletion and erasure: soft delete, hard delete, anonymization, cascading delete behaviors across relationships, and verification steps. This ensures privacy compliance and prevents partial/failed deletions.

Inputs Required

- DLR-01: {{xref:DLR-01}} | OPTIONAL
- DLR-02: {{xref:DLR-02}} | OPTIONAL
- DGP-02: {{xref:DGP-02}} | OPTIONAL

- DGP-04: {{xref:DGP-04}} | OPTIONAL
- DATA-02: {{xref:DATA-02}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Deletion types supported:
    - soft delete (tombstone)
    - hard delete (physical removal)
    - anonymize/pseudonymize (if used)
- Deletion triggers:
    - user request
    - retention expiry
    - admin action
- Procedure definitions (step-by-step) for each deletion type
- Entity deletion matrix (minimum 12 entities):
    - entity_id
    - deletion_type(s) allowed
    - cascade targets (related entities)
    - constraints/locks (legal hold, open invoices, etc.)
    - verification query
    - rollback/undo possibility (soft delete only)

    - audit requirements
- Cross-system deletion scope:
    - caches
    - search indexes
    - analytics/reporting stores
    - backups policy pointer (BDR/STORE)
- Failure handling (partial delete, retries, DLQ)

Optional Fields

- Secure deletion verification methods | OPTIONAL
- Notes | OPTIONAL

Rules

- Hard delete must be final; define irreversibility.
- Cascades must align with DATA-02 relationships; no ad-hoc cascades.
- Legal hold overrides deletion; must be checked first.
- Deletion must include index/cache invalidation.
- Backups: define policy for delete propagation (cannot claim immediate delete from immutable backups; must specify retention window).

Output Format

1) Deletion Types (required)

- Soft delete definition: {{types.soft}}

- Hard delete definition: {{types.hard}}
- Anonymize definition: {{types.anonymize}} | OPTIONAL

2) Triggers (required)

- User-request deletion: {{triggers.user_request}}
- Retention expiry deletion: {{triggers.retention_expiry}}
- Admin deletion: {{triggers.admin}} | OPTIONAL

3) Procedures (required)

Soft Delete Procedure

1. Preconditions: {{proc.soft.preconditions}}
2. Steps: {{proc.soft.steps}}
3. Verification: {{proc.soft.verify}}
4. Undo policy: {{proc.soft.undo_policy}} | OPTIONAL

Hard Delete Procedure

1. Preconditions: {{proc.hard.preconditions}}
2. Steps: {{proc.hard.steps}}
3. Verification: {{proc.hard.verify}}
4. Irreversible statement: {{proc.hard.irreversible}}

Anonymize Procedure (optional)

1. Preconditions: {{proc.anon.preconditions}} | OPTIONAL
2. Steps: {{proc.anon.steps}} | OPTIONAL

3. Verification: {{proc.anon.verify}} | OPTIONAL

4) Entity Deletion Matrix (canonical)

entity_id	allowed_deletion	cascade_targets	constraints	verify_query	undo_possible	audit_required	notes
{{matrix[0].entity}}	{{matrix[0].allowed}}	{{matrix[0].cascade}}	{{matrix[0].constraints}}	{{matrix[0].verify}}	{{matrix[0].undo}}	{{matrix[0].audit}}	{{matrix[0].notes}}
{{matrix[1].entity}}	{{matrix[1].allowed}}	{{matrix[1].cascade}}	{{matrix[1].constraints}}	{{matrix[1].verify}}	{{matrix[1].undo}}	{{matrix[1].audit}}	{{matrix[1].notes}}

5) Cross-System Deletion Scope (required)

- Cache invalidation: {{scope.cache}}
- Search index removal: {{scope.search}}
- Analytics/reporting handling: {{scope.analytics_reporting}}
- Backup policy pointer: {{scope.backups_pointer}} | OPTIONAL

6) Failure Handling (required)

- Partial deletion handling: {{failure.partial}}
- Retry policy pointer: {{xref:ERR-05}} | OPTIONAL
- DLQ/quarantine rule: {{failure.dlq}} | OPTIONAL

Cross-References

- Upstream: {{xref:DLR-02}} | OPTIONAL, {{xref:DATA-02}} | OPTIONAL, {{xref:DGP-04}} | OPTIONAL
- Downstream: {{xref:DLR-04}} | OPTIONAL, {{xref:SRCH-03}} | OPTIONAL, {{xref:CACHE-02}} | OPTIONAL, {{xref:BDR-*}} | OPTIONAL

- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Soft vs hard delete procedures + matrix basics.
- intermediate: Required. Add cascade targets, verification, and cross-system scope.
- advanced: Required. Add failure handling, backups posture, and anonymization (if used).

Unknown Handling

- UNKNOWN_ALLOWED: `anonymization`, `secure_delete_verification`, `notes`, `admin_trigger` (if no admin deletes)
- If backups policy is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
  - required_fields_present == true
  - procedures_present == true
  - entity_matrix_present == true
  - cross_system_scope_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DLR-04

DLR-04 — Legal Holds & Exceptions Policy

Header Block

- template_id: DLR-04
- title: Legal Holds & Exceptions Policy
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-04_Legal_Holds_Exceptions_Policy.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["DLR-02", "COMP-01", "GOVOPS-03"]
- inputs_required: ["DLR-02", "COMP-01", "DGP-02", "GOVOPS-03", "AUDIT-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how legal holds and retention exceptions work: when holds apply, what data is frozen, how deletion is blocked, who can apply/release holds, and how holds are audited. This ensures compliance and prevents accidental deletion.

Inputs Required

- DLR-02: {{xref:DLR-02}} | OPTIONAL
- COMP-01: {{xref:COMP-01}} | OPTIONAL
- DGP-02: {{xref:DGP-02}} | OPTIONAL

- GOVOPS-03: {{xref:GOVOPS-03}} | OPTIONAL
- AUDIT-01: {{xref:AUDIT-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Legal hold definition and scope
- Hold triggers (lawsuit/investigation/compliance request)
- Who can place/release holds (roles, approvals)
- Hold effects:
    - block deletion
    - block anonymization (if required)
    - allow read-only access
    - export rules
- Exceptions policy:
    - when retention can be extended
    - when retention can be shortened (rare)
- Hold tracking fields (hold_id, scope, start/end, reason)
- Audit requirements
- Verification checklist

Optional Fields

- Jurisdiction-specific rules | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Holds override deletion procedures (DLR-03) and retention timers (DLR-02).
- Any hold action must be auditable with reason and approver.
- Hold release must be explicit and recorded.

Output Format

1) Applicability

- applies: {{legal_holds.applies}} (true/false)
- 00_NA (if not applies): {{legal_holds.na_block}} | OPTIONAL

2) Legal Hold Policy (required if applies)

- Definition: {{policy.definition}}
- Scope levels: {{policy.scope_levels}} (user/account/entity/dataset)
- Effects: {{policy.effects}}
- Allowed access during hold: {{policy.access_during_hold}} | OPTIONAL

3) Roles & Approval (required if applies)

- Who can place holds: {{roles.place}}
- Who can release holds: {{roles.release}}

- Approval requirements: {{roles.approval}}
- Required justification fields: {{roles.justification}} | OPTIONAL

4) Exceptions Policy (required if applies)

- Extend retention when: {{exceptions.extend_when}}
- Shorten retention when: {{exceptions.shorten_when}} | OPTIONAL
- Required approvals: {{exceptions.approvals}}
- Time-bound rule: {{exceptions.time_bound}} | OPTIONAL

5) Tracking Fields (required if applies)

field	meaning	required
hold_id	{{fields.hold_id}}	true
scope	{{fields.scope}}	true
reason	{{fields.reason}}	true
start_at	{{fields.start_at}}	true
end_at	{{fields.end_at}}	false

6) Audit Requirements (required if applies)

- Audit events: {{audit.events}}
- Retention of audit logs: {{audit.retention}} | OPTIONAL

7) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DLR-02}} | OPTIONAL, {{xref:COMP-01}} | OPTIONAL
- Downstream: {{xref:DLR-03}} | OPTIONAL, {{xref:ADMIN-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Define triggers + roles + effects.
- advanced: Required if applies. Add exceptions governance and audit rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `jurisdiction_rules`, `notes`, `shorten_retention` (if disallowed)
- If applies == true and roles/approval are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_policy_present == true
    - roles_approval_present == true
    - tracking_fields_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

DLR-05

DLR-05 — Archival & Cold Storage Strategy

Header Block

- template_id: DLR-05
- title: Archival & Cold Storage Strategy
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-05_Archival_Cold_Storage_Strategy.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["DLR-01", "DLR-02", "STORE-01", "RPT-04"]
- inputs_required: ["DLR-01", "DLR-02", "STORE-01", "RPT-04", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how archived data is stored, accessed, and restored: what moves to cold storage, when it moves, how it is indexed (if at all), how access is controlled, and how archival impacts reporting, search, and cost.

Inputs Required

- DLR-01: {{xref:DLR-01}} | OPTIONAL
- DLR-02: {{xref:DLR-02}} | OPTIONAL
- STORE-01: {{xref:STORE-01}} | OPTIONAL

- RPT-04: {{xref:RPT-04}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Archival candidates list (entities/datasets)
- Archival trigger rules (age, inactivity, lifecycle transition)
- Storage tiers and locations (hot/cold) (pointer to STORE)
- Access policy for archived data:
    - who can access
    - how it’s requested (on-demand restore vs direct query)
    - latency expectation
- Indexing/search policy for archived data (SRCH pointer)
- Reporting impact policy (RPT pointer)
- Restore procedure (high level) + verification
- Cost control notes (qualitative)

Optional Fields

- Rehydration policy (bring back to hot) | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Archived data access must respect permissions and privacy.
- Restore must be auditable for sensitive data.
- Search and reporting policies must explicitly state whether archived data is included.

Output Format

1) Applicability

- applies: {{archival.applies}} (true/false)
- 00_NA (if not applies): {{archival.na_block}} | OPTIONAL

2) Archival Candidates (required if applies)

entity_or_dataset	trigger_rule	storage_tier	access_mode	search_included	reporting_included	restore_required	notes
{{candidates[0].name}}	{{candidates[0].trigger}}	{{candidates[0].tier}}	{{candidates[0].access}}	{{candidates[0].search}}	{{candidates[0].reporting}}	{{candidates[0].restore}}	{{candidates[0].notes}}

3) Access Policy (required if applies)

- Who can access: {{access.who}}
- How access happens: {{access.how}}
- Latency expectation: {{access.latency}} | OPTIONAL
- Audit requirement: {{access.audit}} | OPTIONAL

4) Indexing/Search Policy (required if applies)

- Included in search: {{search.included}}
- If excluded, behavior: {{search.excluded_behavior}} | OPTIONAL

- SRCH pointer: {{xref:SRCH-01}} | OPTIONAL

5) Reporting Policy (required if applies)

- Included in reporting: {{reporting.included}}
- Rollup/snapshot behavior: {{reporting.rollup_behavior}} | OPTIONAL
- RPT pointer: {{xref:RPT-04}} | OPTIONAL

6) Restore Procedure (required if applies)

1. Request/approval: {{restore.request}}
2. Retrieve/rehydrate: {{restore.retrieve}}
3. Verify: {{restore.verify}}
4. Expiry/back to cold (if temporary): {{restore.expiry}} | OPTIONAL

7) Cost Controls (required if applies)

- Primary cost drivers: {{cost.drivers}}
- Cost control levers: {{cost.levers}}

Cross-References

- Upstream: {{xref:STORE-01}} | OPTIONAL, {{xref:DLR-02}} | OPTIONAL
- Downstream: {{xref:STORE-02}} | OPTIONAL, {{xref:STORE-03}} | OPTIONAL, {{xref:SRCH-03}} | OPTIONAL, {{xref:RPT-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Candidates + access policy + restore procedure.
- advanced: Required if applies. Add audit requirements and reporting/search inclusion rules.

Unknown Handling

- UNKNOWN_ALLOWED: `rehydration_policy`, `notes`, `latency_expectation`, `audit_requirement`
- If applies == true and restore procedure is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_candidates_present == true
    - access_policy_present == true
    - restore_procedure_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DLR-06

DLR-06 — Data Minimization Rules (collect/store only what’s needed)

Header Block

- template_id: DLR-06
- title: Data Minimization Rules (collect/store only what’s needed)
- type: data_lifecycle_retention
- template_version: 1.0.0
- output_path: 10_app/data_lifecycle/DLR-06_Data_Minimization_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.DLR
- upstream_dependencies: ["PRD-06", "DGP-01", "DGP-02"]
- inputs_required: ["PRD-06", "DGP-01", "DGP-02", "FORM-01", "API-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define rules to minimize collected and stored data: which fields are necessary, which are optional, when data should be avoided, and what alternatives exist. This reduces privacy risk and storage cost while supporting product requirements.

Inputs Required

- PRD-06: {{xref:PRD-06}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- DGP-02: {{xref:DGP-02}} | OPTIONAL

- FORM-01: {{xref:FORM-01}} | OPTIONAL
- API-02: {{xref:API-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Minimization principles (explicit)
- “Do not collect” list (fields/data types)
- “Collect only when needed” list (conditional fields with triggers)
- Required justification rules for high-PII fields
- Storage minimization rules:
    - avoid duplication
    - store derived values only when needed
    - redact/trim logs
- UX constraints:
    - optional fields must be optional in UX
    - explain why requested (copy pointer)
- Verification checklist

Optional Fields

- Field-level minimization matrix | OPTIONAL
- Notes | OPTIONAL

Rules

- If a field is optional in product logic, it must not be required in forms.
- High-PII fields require explicit business justification and retention policy.
- Logs/telemetry must follow minimization; do not collect full payloads by default.
- “Just in case” collection is disallowed.

Output Format

1) Principles (required)

- {{principles[0]}}
- {{principles[1]}}
- {{principles[2]}} | OPTIONAL

2) Do Not Collect (required)

- {{do_not_collect[0]}}
- {{do_not_collect[1]}} | OPTIONAL

3) Collect Only When Needed (required)

field_or_data	collect_when	purpose	retention_pointer	notes
{{conditional[0].field}}	{{conditional[0].when}}	{{conditional[0].purpose}}	{{conditional[0].retention}}	{{conditional[0].notes}}

4) High-PII Justification Rules (required)

- Required justification fields: {{pii.justification_fields}}
- Approval requirement: {{pii.approval}} | OPTIONAL
- Required retention mapping: {{pii.retention_mapping}}

5) Storage & Logging Minimization (required)

- Duplication avoidance: {{storage.dup_avoid}}
- Derived storage policy: {{storage.derived_policy}} | OPTIONAL
- Logging redaction rule: {{storage.logging_redaction}}

6) UX Constraints (required)

- Optional in UX rule: {{ux.optional_rule}}
- Explain-why copy pointer: {{xref:CDX-02}} | OPTIONAL

7) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DGP-01}} | OPTIONAL, {{xref:PRD-06}} | OPTIONAL
- Downstream: {{xref:DLR-02}} | OPTIONAL, {{xref:DQV-03}} | OPTIONAL, {{xref:OBS-01}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Principles + do-not-collect list.
- intermediate: Required. Add conditional collection table and logging minimization.
- advanced: Required. Add high-PII justification and approval mapping.

Unknown Handling

- UNKNOWN_ALLOWED: `field_level_matrix`, `notes`, `approval_requirement`
- If do-not-collect list is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DLR
- Pass conditions:
    - required_fields_present == true
    - principles_present == true
    - do_not_collect_present == true
    - verification_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Data Governance & Lineage (DGL)

Data Governance & Lineage (DGL)

- DGL-01 Data Ownership Map (owner per entity/dataset)
- DGL-02 Lineage Map (sources → transforms → sinks)
- DGL-03 Transformation Rules Catalog (ETL/ELT, normalization)
- DGL-04 Access Controls for Data (who can read/write/export)
- DGL-05 Auditability Requirements (what must be traceable)
- DGL-06 Data Catalog / Dictionary (datasets, meaning, sensitivity)

DGL-01

DGL-01 — Data Ownership Map (owner per entity/dataset)

Header Block

- template_id: DGL-01
- title: Data Ownership Map (owner per entity/dataset)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-01_Data_Ownership_Map.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["DATA-01", "ARC-01", "STK-03"]
- inputs_required: ["DATA-01", "ARC-01", "STK-03", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define who “owns” each entity/dataset: the accountable party for correctness, schema changes, access controls, retention policy alignment, and incident response. This prevents orphaned data with unclear decision rights.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- ARC-01: {{xref:ARC-01}} | OPTIONAL
- STK-03: {{xref:STK-03}} | OPTIONAL

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Ownership registry entries (minimum: all DATA-01 entities + key derived datasets)
- For each entry:
    - entity_id or dataset_id
    - owner_role/team
    - steward (optional second owner)
    - change approver (who approves schema changes)
    - access approver (who approves access/export)
    - oncall/escalation contact (role, not person) | OPTIONAL
    - sensitivity class (PII level)
    - criticality (P0/P1/P2)
- Ownership change procedure

Optional Fields

- Data product classification | OPTIONAL
- Notes | OPTIONAL

Rules

- Every entity must have exactly one primary owner.

- High-sensitivity entities require explicit access approver.
- Ownership must align with boundary ownership (ARC-01); if mismatch, justify.
- Ownership changes must be auditable.

Output Format

1) Ownership Registry (canonical)

id	kind (entity/dataset)	owner	steward	schema_approver	access_approver	escalation_role	sensitivity	criticality	notes
{{owners[0].id}}	{{owners[0].kind}}	{{owners[0].owner}}	{{owners[0].steward}}	{{owners[0].schema_approver}}	{{owners[0].access_approver}}	{{owners[0].escalation}}	{{owners[0].sensitivity}}	{{owners[0].criticality}}	{{owners[0].notes}}
{{owners[1].id}}	{{owners[1].kind}}	{{owners[1].owner}}	{{owners[1].steward}}	{{owners[1].schema_approver}}	{{owners[1].access_approver}}	{{owners[1].escalation}}	{{owners[1].sensitivity}}	{{owners[1].criticality}}	{{owners[1].notes}}

2) Ownership Change Procedure (required)

- How to request change: {{change.request}}
- Required approvals: {{change.approvals}}
- Audit/log requirement: {{change.audit}} | OPTIONAL

Cross-References

- Upstream: {{xref:ARC-01}} | OPTIONAL, {{xref:STK-03}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:DGL-04}} | OPTIONAL, {{xref:DLR-02}} | OPTIONAL, {{xref:DQV-04}} | OPTIONAL

- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Ownership registry with primary owners.
- intermediate: Required. Add approvers and sensitivity/criticality.
- advanced: Required. Add escalation role and change procedure rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `steward`, `escalation_role`, `data_product_classification`, `notes`
- If any entity lacks primary owner → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL
- Pass conditions:
  - required_fields_present == true
  - ownership_complete_for_entities == true
  - exactly_one_primary_owner_each == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DGL-02

DGL-01 — Data Ownership Map (owner per entity/dataset)

Header Block

- template_id: DGL-01
- title: Data Ownership Map (owner per entity/dataset)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-01_Data_Ownership_Map.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["DATA-01", "ARC-01", "STK-03"]
- inputs_required: ["DATA-01", "ARC-01", "STK-03", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define who “owns” each entity/dataset: the accountable party for correctness, schema changes, access controls, retention policy alignment, and incident response. This prevents orphaned data with unclear decision rights.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- ARC-01: {{xref:ARC-01}} | OPTIONAL
- STK-03: {{xref:STK-03}} | OPTIONAL

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Ownership registry entries (minimum: all DATA-01 entities + key derived datasets)
- For each entry:
    - entity_id or dataset_id
    - owner_role/team
    - steward (optional second owner)
    - change approver (who approves schema changes)
    - access approver (who approves access/export)
    - oncall/escalation contact (role, not person) | OPTIONAL
    - sensitivity class (PII level)
    - criticality (P0/P1/P2)
- Ownership change procedure

Optional Fields

- Data product classification | OPTIONAL
- Notes | OPTIONAL

Rules

- Every entity must have exactly one primary owner.

- High-sensitivity entities require explicit access approver.
- Ownership must align with boundary ownership (ARC-01); if mismatch, justify.
- Ownership changes must be auditable.

Output Format

1) Ownership Registry (canonical)

id	kind (entity/dataset)	owner	steward	schema_approver	access_approver	escalation_role	sensitivity	criticality	notes
{{owners[0].id}}	{{owners[0].kind}}	{{owners[0].owner}}	{{owners[0].steward}}	{{owners[0].schema_approver}}	{{owners[0].access_approver}}	{{owners[0].escalation}}	{{owners[0].sensitivity}}	{{owners[0].criticality}}	{{owners[0].notes}}
{{owners[1].id}}	{{owners[1].kind}}	{{owners[1].owner}}	{{owners[1].steward}}	{{owners[1].schema_approver}}	{{owners[1].access_approver}}	{{owners[1].escalation}}	{{owners[1].sensitivity}}	{{owners[1].criticality}}	{{owners[1].notes}}

2) Ownership Change Procedure (required)

- How to request change: {{change.request}}
- Required approvals: {{change.approvals}}
- Audit/log requirement: {{change.audit}} | OPTIONAL

Cross-References

- Upstream: {{xref:ARC-01}} | OPTIONAL, {{xref:STK-03}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:DGL-04}} | OPTIONAL, {{xref:DLR-02}} | OPTIONAL, {{xref:DQV-04}} | OPTIONAL

- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Ownership registry with primary owners.
- intermediate: Required. Add approvers and sensitivity/criticality.
- advanced: Required. Add escalation role and change procedure rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `steward`, `escalation_role`, `data_product_classification`, `notes`
- If any entity lacks primary owner → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL
- Pass conditions:
  - required_fields_present == true
  - ownership_complete_for_entities == true
  - exactly_one_primary_owner_each == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DGL-03

DGL-03 — Transformation Rules Catalog (ETL/ELT, normalization)

Header Block

- template_id: DGL-03
- title: Transformation Rules Catalog (ETL/ELT, normalization)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-03_Transformation_Rules_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["DGL-02", "SIC-04", "DLR-06"]
- inputs_required: ["DGL-02", "SIC-04", "DLR-06", "DQV-01", "ERR-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the canonical catalog of transformation rules applied to data as it moves through the system: mapping, normalization, enrichment, redaction, aggregation, and formatting rules. This prevents undocumented transforms and supports repeatability and audits.

Inputs Required

- DGL-02: {{xref:DGL-02}} | OPTIONAL
- SIC-04: {{xref:SIC-04}} | OPTIONAL
- DLR-06: {{xref:DLR-06}} | OPTIONAL

- DQV-01: {{xref:DQV-01}} | OPTIONAL
- ERR-02: {{xref:ERR-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Transformation rule catalog (minimum 20 rules for non-trivial systems; justify if smaller)
- For each rule:
    - rule_id
    - name
    - category (normalize/map/enrich/aggregate/redact)
    - inputs (fields/entities)
    - output (fields/entities)
    - transform logic description (deterministic)
    - validation rule/predicate
    - failure behavior (reject/quarantine/default)
    - reason_code on failure (if reject)
    - owner
    - applied_in (transform step IDs from DGL-02)
    - data minimization impact (what is removed/trimmed) | OPTIONAL

Optional Fields

- Example input/output pairs | OPTIONAL

- Notes | OPTIONAL

Rules

- Transform logic must be deterministic and described without code.
- Any “default” behavior must be explicit and justified.
- Redaction transforms must align with privacy policy and be irreversible where required.
- Every reject path must map to a reason_code.

Output Format

Transformation Rules (canonical)

rule_id	category	name	inputs	outputs	logic	validation	failure_behavior	reason_code	applied_in_steps	owner	notes
tr_rule_01	{{rules[0].category}}	{{rules[0].name}}	{{rules[0].inputs}}	{{rules[0].outputs}}	{{rules[0].logic}}	{{rules[0].validation}}	{{rules[0].failure}}	{{rules[0].rc}}	{{rules[0].steps}}	{{rules[0].owner}}	{{rules[0].notes}}
tr_rule_02	{{rules[1].category}}	{{rules[1].name}}	{{rules[1].inputs}}	{{rules[1].outputs}}	{{rules[1].logic}}	{{rules[1].validation}}	{{rules[1].failure}}	{{rules[1].rc}}	{{rules[1].steps}}	{{rules[1].owner}}	{{rules[1].notes}}

Example Pairs (optional)

- Example input: {{examples[0].input}} | OPTIONAL
- Example output: {{examples[0].output}} | OPTIONAL

Cross-References

- Upstream: {{xref:DGL-02}} | OPTIONAL, {{xref:SIC-04}} | OPTIONAL, {{xref:DLR-06}} | OPTIONAL
- Downstream: {{xref:DQV-02}} | OPTIONAL, {{xref:PIPE-04}} | OPTIONAL, {{xref:BI-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. 20 rule entries with failure behaviors.
- advanced: Required if applies. Add applied_in_steps mapping and reason codes.

Unknown Handling

- UNKNOWN_ALLOWED: `example_pairs`, `notes`, `minimization_impact`
- If any reject path lacks reason_code → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL
- Pass conditions:
    - required_fields_present == true
    - rules_count >= 20 (or justified)
    - failure_behaviors_defined == true
    - reject_paths_have_reason_codes == true
    - placeholder_resolution == true

    - no_unapproved_unknowns == true

DGL-04 — Access Controls for Data (who can read/write/export)

Header Block

- template_id: DGL-04
- title: Access Controls for Data (who can read/write/export)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-04_Access_Controls_for_Data.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["PMAD-02", "DGP-01", "DGL-01"]
- inputs_required: ["PMAD-02", "DGP-01", "DGL-01", "COMP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the data-layer access control rules: who can read/write/export which entities/datasets, what constraints apply for sensitive data, and what approval/audit requirements exist.

Inputs Required

- PMAD-02: {{xref:PMAD-02}} | OPTIONAL

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- DGL-01: {{xref:DGL-01}} | OPTIONAL
- COMP-01: {{xref:COMP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Access control matrix (minimum 20 entries)
- For each entry:
    - entity_id/dataset_id
    - action (read/write/export/delete)
    - allowed roles (role_id)
    - constraints (tenant/ownership/status/legal hold)
    - approval required (yes/no + who)
    - audit required (yes/no + event type)
    - masking/redaction rules (for sensitive fields)
- Export policy (bulk export restrictions)
- Break-glass rules pointer (PMAD-05)
- Verification checklist

Optional Fields

- Data masking patterns | OPTIONAL
- Notes | OPTIONAL

Rules

- Export is treated as privileged for high-PII datasets.
- Masking/redaction must be explicit for sensitive fields.
- Access rules must align with PMAD policy rules; do not create contradictory rules.
- All access to sensitive datasets must be auditable.

Output Format

1) Access Control Matrix (canonical)

id	kind	action	allowed_roles	constraints	approval	audit_event	masking_rules	notes
{{matrix[0].id}}	{{matrix[0].kind}}	{{matrix[0].action}}	{{matrix[0].roles}}	{{matrix[0].constraints}}	{{matrix[0].approval}}	{{matrix[0].audit}}	{{matrix[0].masking}}	{{matrix[0].notes}}
{{matrix[1].id}}	{{matrix[1].kind}}	{{matrix[1].action}}	{{matrix[1].roles}}	{{matrix[1].constraints}}	{{matrix[1].approval}}	{{matrix[1].audit}}	{{matrix[1].masking}}	{{matrix[1].notes}}

2) Export Policy (required)

- Export allowed when: {{export.allowed_when}}
- Export denied when: {{export.denied_when}}
- Approval requirement: {{export.approval}}
- Rate limits for export: {{export.ratelimits}} | OPTIONAL

3) Break-Glass Pointer (required)

- PMAD-05 pointer: {{xref:PMAD-05}} | OPTIONAL

4) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PMAD-02}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:DGL-05}} | OPTIONAL, {{xref:DLR-02}} | OPTIONAL, {{xref:ADMIN-02}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Matrix with read/write basics and audit flags.
- intermediate: Required. Add export rules and masking for sensitive data.
- advanced: Required. Add approval governance and verification checklist rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `masking_patterns`, `notes`, `ratelimits`
- If any sensitive dataset lacks audit_event requirement → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL

- Pass conditions:
    - required_fields_present == true
    - matrix_count >= 20
    - export_policy_present == true
    - sensitive_access_audited == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DGL-04

DGL-05 — Auditability Requirements (what must be traceable)

Header Block

- template_id: DGL-05
- title: Auditability Requirements (what must be traceable)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-05_Auditability_Requirements.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["PMAD-06", "AUDIT-01", "DGL-04"]
- inputs_required: ["PMAD-06", "AUDIT-01", "DGL-04", "DGP-01", "OBS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define what data actions must be traceable end-to-end: create/update/delete, access, exports, transformations, and administrative repairs. This consolidates audit requirements into a concrete checklist and event catalog.

Inputs Required

- PMAD-06: {{xref:PMAD-06}} | OPTIONAL
- AUDIT-01: {{xref:AUDIT-01}} | OPTIONAL
- DGL-04: {{xref:DGL-04}} | OPTIONAL

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- OBS-01: {{xref:OBS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Traceability event catalog (minimum 25 events)
- For each event:
    - event_id
    - category (data_write/data_read/data_export/transform/admin_repair)
    - entity_id/dataset_id
    - action
    - who (actor types: user/service/admin)
    - required fields (resource identifiers, before/after pointers, reason_code)
    - redaction rules
    - retention window for audit record
    - access controls for audit viewing
- Required trace linkage:
    - correlation_id / trace_id propagation rules
- Verification checklist

Optional Fields

- Evidence pack requirements for compliance audits | OPTIONAL

- Notes | OPTIONAL

Rules

- Any export and any privileged data repair must be auditable.
- Before/after must be stored in a safe form (redacted) or as pointers to snapshots.
- Audit logs must be tamper-evident or protected (system pointer).
- Viewing audit logs is itself an auditable action.

Output Format

1) Audit Event Catalog (canonical)

event_id	category	entity_or_dataset	action	actor_types	required_fields	redaction	retention	view_roles	notes
aud_01	{{events[0].category}}	{{events[0].target}}	{{events[0].action}}	{{events[0].actors}}	{{events[0].fields}}	{{events[0].redaction}}	{{events[0].retention}}	{{events[0].view_roles}}	{{events[0].notes}}
aud_02	{{events[1].category}}	{{events[1].target}}	{{events[1].action}}	{{events[1].actors}}	{{events[1].fields}}	{{events[1].redaction}}	{{events[1].retention}}	{{events[1].view_roles}}	{{events[1].notes}}

2) Trace Linkage Rules (required)

- correlation_id required: {{trace.correlation_required}}
- trace_id propagation: {{trace.propagation}}
- async propagation: {{trace.async}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PMAD-06}} | OPTIONAL, {{xref:AUDIT-01}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL
- Downstream: {{xref:COMP-02}} | OPTIONAL, {{xref:GOVOPS-05}} | OPTIONAL, {{xref:ADMIN-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Catalog with categories and required fields.
- intermediate: Required. Add retention/view roles and trace linkage rules.
- advanced: Required. Add evidence pack requirements and tamper-evidence pointers.

Unknown Handling

- UNKNOWN_ALLOWED: `evidence_pack_requirements`, `notes`, `async_propagation`
- If export events are missing from catalog → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL

- Pass conditions:
    - required_fields_present == true
    - events_count >= 25
    - export_and_admin_repairs_audited == true
    - trace_linkage_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DGL-05

DGL-06 — Data Catalog / Dictionary (datasets, meaning, sensitivity)

Header Block

- template_id: DGL-06
- title: Data Catalog / Dictionary (datasets, meaning, sensitivity)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-06_Data_Catalog_Dictionary.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["DATA-01", "DMG-01", "DGP-01"]
- inputs_required: ["DATA-01", "DMG-01", "DGP-01", "DGL-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create a human- and machine-friendly dictionary of datasets/entities: what they mean, what they contain, how sensitive they are, who owns them, and what their retention and access rules are. This is the “catalog layer” for data governance.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DMG-01: {{xref:DMG-01}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL

- DGL-01: {{xref:DGL-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Catalog entries (minimum: all DATA-01 entities + key derived/read models + reporting datasets)
- For each entry:
    - id (entity_id/dataset_id/read_model_id)
    - name
    - description (plain language)
    - owner (DGL-01 pointer)
    - sensitivity class (PII level)
    - key fields (top 10)
    - retention pointer (DLR-02)
    - access pointer (DGL-04)
    - lineage pointer (DGL-02) | OPTIONAL
    - typical use cases (API/search/reporting)
    - common pitfalls (gotchas)
- Verification checklist

Optional Fields

- Data quality score pointer (DQV) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align terms with DMG glossary.
- Sensitivity must be explicit and consistent with DGP classification.
- Catalog must include derived/read models and reporting datasets to prevent “shadow data.”
- Catalog entries must remain stable and versioned.

Output Format

Data Catalog Entries (canonical)

id	name	description	owner	sensitivity	key_fields	retention_ref	access_ref	lineage_ref	use_cases	pitfalls	notes
{{catalog[0].id}}	{{catalog[0].name}}	{{catalog[0].desc}}	{{catalog[0].owner}}	{{catalog[0].sens}}	{{catalog[0].key_fields}}	{{catalog[0].retention}}	{{catalog[0].access}}	{{catalog[0].lineage}}	{{catalog[0].use_cases}}	{{catalog[0].pitfalls}}	{{catalog[0].notes}}
{{catalog[1].id}}	{{catalog[1].name}}	{{catalog[1].desc}}	{{catalog[1].owner}}	{{catalog[1].sens}}	{{catalog[1].key_fields}}	{{catalog[1].retention}}	{{catalog[1].access}}	{{catalog[1].lineage}}	{{catalog[1].use_cases}}	{{catalog[1].pitfalls}}	{{catalog[1].notes}}

Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-01}} | OPTIONAL, {{xref:DGL-01}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:DQV-02}} | OPTIONAL, {{xref:RPT-01}} | OPTIONAL, {{xref:SRCH-01}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Entries with name/description/sensitivity/owner.
- intermediate: Required. Add retention/access pointers and key fields.
- advanced: Required. Add lineage pointers and pitfalls/use-cases rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `dq_score_pointer`, `notes`, `lineage_ref` (if no lineage tracking yet)
- If any entity lacks sensitivity classification → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL
- Pass conditions:
    - required_fields_present == true
    - catalog_complete_for_entities == true
    - sensitivity_present == true
    - retention_and_access_refs_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

DGL-06

DGL-06 — Data Catalog / Dictionary (datasets, meaning, sensitivity)

Header Block

- template_id: DGL-06
- title: Data Catalog / Dictionary (datasets, meaning, sensitivity)
- type: data_governance_lineage
- template_version: 1.0.0
- output_path: 10_app/data_governance/DGL-06_Data_Catalog_Dictionary.md
- compliance_gate_id: TMP-05.PRIMARY.DGL
- upstream_dependencies: ["DATA-01", "DMG-01", "DGP-01"]
- inputs_required: ["DATA-01", "DMG-01", "DGP-01", "DGL-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create a human- and machine-friendly dictionary of datasets/entities: what they mean, what they contain, how sensitive they are, who owns them, and what their retention and access rules are. This is the “catalog layer” for data governance.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DMG-01: {{xref:DMG-01}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL

- DGL-01: {{xref:DGL-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Catalog entries (minimum: all DATA-01 entities + key derived/read models + reporting datasets)
- For each entry:
    - id (entity_id/dataset_id/read_model_id)
    - name
    - description (plain language)
    - owner (DGL-01 pointer)
    - sensitivity class (PII level)
    - key fields (top 10)
    - retention pointer (DLR-02)
    - access pointer (DGL-04)
    - lineage pointer (DGL-02) | OPTIONAL
    - typical use cases (API/search/reporting)
    - common pitfalls (gotchas)
- Verification checklist

Optional Fields

- Data quality score pointer (DQV) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align terms with DMG glossary.
- Sensitivity must be explicit and consistent with DGP classification.
- Catalog must include derived/read models and reporting datasets to prevent “shadow data.”
- Catalog entries must remain stable and versioned.

Output Format

Data Catalog Entries (canonical)

id	name	description	owner	sensitivity	key_fields	retention_ref	access_ref	lineage_ref	use_cases	pitfalls	notes
{{catalog[0].id}}	{{catalog[0].name}}	{{catalog[0].desc}}	{{catalog[0].owner}}	{{catalog[0].sens}}	{{catalog[0].key_fields}}	{{catalog[0].retention}}	{{catalog[0].access}}	{{catalog[0].lineage}}	{{catalog[0].use_cases}}	{{catalog[0].pitfalls}}	{{catalog[0].notes}}
{{catalog[1].id}}	{{catalog[1].name}}	{{catalog[1].desc}}	{{catalog[1].owner}}	{{catalog[1].sens}}	{{catalog[1].key_fields}}	{{catalog[1].retention}}	{{catalog[1].access}}	{{catalog[1].lineage}}	{{catalog[1].use_cases}}	{{catalog[1].pitfalls}}	{{catalog[1].notes}}

Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-01}} | OPTIONAL, {{xref:DGL-01}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:DQV-02}} | OPTIONAL, {{xref:RPT-01}} | OPTIONAL, {{xref:SRCH-01}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Entries with name/description/sensitivity/owner.
- intermediate: Required. Add retention/access pointers and key fields.
- advanced: Required. Add lineage pointers and pitfalls/use-cases rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `dq_score_pointer`, `notes`, `lineage_ref` (if no lineage tracking yet)
- If any entity lacks sensitivity classification → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DGL
- Pass conditions:
    - required_fields_present == true
    - catalog_complete_for_entities == true
    - sensitivity_present == true
    - retention_and_access_refs_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

Data Quality & Validation (DQV

Data Quality & Validation (DQV)

- DQV-01 Validation Policy (schema vs semantic)
- DQV-02 Data Quality Checks Catalog (rules by entity)
- DQV-03 Input Validation Rules (API/forms → entities)
- DQV-04 Consistency & Integrity Monitoring (drift, anomalies)
- DQV-05 Bad Data Handling (quarantine, repair, backfill)
- DQV-06 DQ Test & Verification Plan

DQV-01

DQV-01 — Validation Policy (schema vs semantic)

Header Block

- template_id: DQV-01
- title: Validation Policy (schema vs semantic)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-01_Validation_Policy.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DATA-01", "BRP-01", "ERR-01"]
- inputs_required: ["DATA-01", "BRP-01", "ERR-01", "API-02", "FORM-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the system's validation policy: what is validated at schema level vs semantic level, where validation happens (client/server/DB), how failures are handled, and how validation ties into error taxonomy and reason codes.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- ERR-01: {{xref:ERR-01}} | OPTIONAL

- API-02: {{xref:API-02}} | OPTIONAL
- FORM-01: {{xref:FORM-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Definitions:
    - schema validation (shape/type/required)
    - semantic validation (cross-field/business invariants)
- Enforcement points:
    - client (forms)
    - server (API boundary)
    - DB (constraints)
- Validation responsibilities matrix:
    - what must be validated where (minimum 12 rules)
- Failure handling rules:
    - reject vs quarantine vs default
    - error_class assignment
    - reason_code policy (ERR-02)
- Unknown/extra fields policy:
    - strict vs permissive parsing
- Performance rules (avoid expensive validation in hot paths)
- Verification checklist

Optional Fields

- Streaming/batch validation notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Client validation improves UX but is non-authoritative; server must enforce.
- Semantic validation must reference canonical rules (DMG-03/BRP).
- Validation failures must map to reason codes where user-visible.
- Unknown fields handling must be explicit and consistent across endpoints.

Output Format

1) Definitions (required)

- Schema validation: {{defs.schema}}
- Semantic validation: {{defs.semantic}}

2) Enforcement Points (required)

- Client: {{enforcement.client}}
- Server: {{enforcement.server}}
- DB: {{enforcement.db}}

3) Responsibilities Matrix (required)

rule_id	validation_type	enforced_at	applies_to	description	failure_behavior	error_class	reason_code	notes

v_01	schema	server	{{matrix[0].applies_to}}	{{matrix[0].desc}}	{{matrix[0].failure}}	{{matrix[0].class}}	{{matrix[0].rc}}	{{matrix[0].notes}}
v_02	semantic	server	{{matrix[1].applies_to}}	{{matrix[1].desc}}	{{matrix[1].failure}}	{{matrix[1].class}}	{{matrix[1].rc}}	{{matrix[1].notes}}

4) Unknown/Extra Fields Policy (required)

- Input strictness: {{unknown.strictness}}
- Extra fields behavior: {{unknown.extra_fields}}
- Logging policy: {{unknown.logging}} | OPTIONAL

5) Performance Rules (required)

- Hot path rule: {{perf.hot_path}}
- Batch validation rule: {{perf.batch}} | OPTIONAL

6) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-01}} | OPTIONAL, {{xref:ERR-01}} | OPTIONAL, {{xref:BRP-01}} | OPTIONAL
- Downstream: {{xref:DQV-02}}, {{xref:DQV-03}}, {{xref:DATA-06}} | OPTIONAL, {{xref:ERR-04}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Definitions + enforcement points.
- intermediate: Required. Add responsibilities matrix and unknown fields policy.
- advanced: Required. Add failure behavior mapping and performance rules.

Unknown Handling

- UNKNOWN_ALLOWED: `streaming_batch_notes`, `notes`, `logging_policy`
- If server enforcement is not specified → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
    - required_fields_present == true
    - definitions_present == true
    - enforcement_points_present == true
    - responsibilities_matrix_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DQV-02

DQV-02 — Data Quality Checks Catalog (rules by entity)

Header Block

- template_id: DQV-02
- title: Data Quality Checks Catalog (rules by entity)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-02_Data_Quality_Checks_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DATA-01", "DATA-03", "DQV-01"]
- inputs_required: ["DATA-01", "DATA-03", "DQV-01", "ERR-02", "OBS-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the catalog of data quality checks that continuously validate correctness, completeness, and integrity of stored data. These checks detect drift, corruption, and broken invariants over time.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-03: {{xref:DATA-03}} | OPTIONAL
- DQV-01: {{xref:DQV-01}} | OPTIONAL

- ERR-02: {{xref:ERR-02}} | OPTIONAL
- OBS-02: {{xref:OBS-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- DQ check catalog (minimum 25 checks)
- For each check:
    - dq_id
    - entity_id/dataset_id
    - check_type (completeness/validity/uniqueness/referential/freshness)
    - rule description (predicate)
    - threshold (allowed failure rate)
    - schedule (cron/streaming/on-write)
    - owner
    - alerting severity
    - failure action (alert/quarantine/stop pipeline)
    - reason_code (if user-impacting) | OPTIONAL
    - metric name (for tracking)
- Reporting format (DQ report contents)

Optional Fields

- Auto-repair strategies | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Each check must be measurable and produce a metric.
- Thresholds must be explicit; “should be good” is not allowed.
- Any “stop pipeline” action must have an escalation path.

Output Format

1) Applicability

- applies: {{dq.applies}} (true/false)
- 00_NA (if not applies): {{dq.na_block}} | OPTIONAL

2) DQ Checks Catalog (canonical)

dq_id	target	check_type	rule	threshold	schedule	owner	severity	action	metric	notes
dq_01	{{checks[0].target}}	{{checks[0].type}}	{{checks[0].rule}}	{{checks[0].threshold}}	{{checks[0].schedule}}	{{checks[0].owner}}	{{checks[0].severity}}	{{checks[0].action}}	{{checks[0].metric}}	{{checks[0].notes}}
dq_	{{checks[1].target}}	{{checks[1].type}}	{{checks[1].rule}}	{{checks[1].threshold}}	{{checks[1].schedule}}	{{checks[1].owner}}	{{checks[1].severity}}	{{checks[1].action}}	{{checks[1].metric}}	{{checks[1].notes}}

0
2

3) DQ Report Format (required if applies)

- Report fields: {{report.fields}}
- Aggregation windows: {{report.windows}} | OPTIONAL
- Storage location pointer: {{report.storage}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-01}} | OPTIONAL, {{xref:DATA-03}} | OPTIONAL
- Downstream: {{xref:DQV-04}}, {{xref:DQV-05}}, {{xref:ALRT-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. 25 checks with thresholds and schedules.
- advanced: Required if applies. Add failure actions, alerting severity, and report storage.

Unknown Handling

- UNKNOWN_ALLOWED: `auto_repair_strategies`, `notes`, `report_storage_pointer`
- If applies == true and thresholds are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_checks_count >= 25
    - thresholds_present == true
    - schedules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DQV-03

DQV-02 — Data Quality Checks Catalog (rules by entity)

Header Block

- template_id: DQV-02
- title: Data Quality Checks Catalog (rules by entity)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-02_Data_Quality_Checks_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DATA-01", "DATA-03", "DQV-01"]
- inputs_required: ["DATA-01", "DATA-03", "DQV-01", "ERR-02", "OBS-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the catalog of data quality checks that continuously validate correctness, completeness, and integrity of stored data. These checks detect drift, corruption, and broken invariants over time.

Inputs Required

- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-03: {{xref:DATA-03}} | OPTIONAL
- DQV-01: {{xref:DQV-01}} | OPTIONAL

- ERR-02: {{xref:ERR-02}} | OPTIONAL
- OBS-02: {{xref:OBS-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- DQ check catalog (minimum 25 checks)
- For each check:
    - dq_id
    - entity_id/dataset_id
    - check_type (completeness/validity/uniqueness/referential/freshness)
    - rule description (predicate)
    - threshold (allowed failure rate)
    - schedule (cron/streaming/on-write)
    - owner
    - alerting severity
    - failure action (alert/quarantine/stop pipeline)
    - reason_code (if user-impacting) | OPTIONAL
    - metric name (for tracking)
- Reporting format (DQ report contents)

Optional Fields

- Auto-repair strategies | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Each check must be measurable and produce a metric.
- Thresholds must be explicit; “should be good” is not allowed.
- Any “stop pipeline” action must have an escalation path.

Output Format

1) Applicability

- applies: {{dq.applies}} (true/false)
- 00_NA (if not applies): {{dq.na_block}} | OPTIONAL

2) DQ Checks Catalog (canonical)

dq_id	target	check_type	rule	threshold	schedule	owner	severity	action	metric	notes
dq_01	{{checks[0].target}}	{{checks[0].type}}	{{checks[0].rule}}	{{checks[0].threshold}}	{{checks[0].schedule}}	{{checks[0].owner}}	{{checks[0].severity}}	{{checks[0].action}}	{{checks[0].metric}}	{{checks[0].notes}}
dq_	{{checks[1].target}}	{{checks[1].type}}	{{checks[1].rule}}	{{checks[1].threshold}}	{{checks[1].schedule}}	{{checks[1].owner}}	{{checks[1].severity}}	{{checks[1].action}}	{{checks[1].metric}}	{{checks[1].notes}}

0
2

3) DQ Report Format (required if applies)

- Report fields: {{report.fields}}
- Aggregation windows: {{report.windows}} | OPTIONAL
- Storage location pointer: {{report.storage}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-01}} | OPTIONAL, {{xref:DATA-03}} | OPTIONAL
- Downstream: {{xref:DQV-04}}, {{xref:DQV-05}}, {{xref:ALRT-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. 25 checks with thresholds and schedules.
- advanced: Required if applies. Add failure actions, alerting severity, and report storage.

Unknown Handling

- UNKNOWN_ALLOWED: `auto_repair_strategies`, `notes`, `report_storage_pointer`
- If applies == true and thresholds are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_checks_count >= 25
    - thresholds_present == true
    - schedules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DQV-04

DQV-04 — Consistency & Integrity Monitoring (drift, anomalies)

Header Block

- template_id: DQV-04
- title: Consistency & Integrity Monitoring (drift, anomalies)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-04_Consistency_Integrity_Monitoring.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DQV-02", "OBS-02", "ALRT-01"]
- inputs_required: ["DQV-02", "OBS-02", "ALRT-01", "RELIA-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how the system monitors data consistency and integrity over time: drift detection, anomaly detection, referential integrity monitoring, and alerting/escalation when data quality degrades.

Inputs Required

- DQV-02: {{xref:DQV-02}} | OPTIONAL
- OBS-02: {{xref:OBS-02}} | OPTIONAL
- ALRT-01: {{xref:ALRT-01}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Monitoring signals catalog (minimum 12 signals)
- For each signal:
    - signal_id
    - target entity/dataset
    - signal type (drift/anomaly/integrity/freshness)
    - metric definition
    - baseline expectation
    - thresholds (warning/critical)
    - window
    - detection frequency
    - alert routing (who gets paged/ticket)
    - failure action (investigate/backfill/pause pipeline)
- Escalation process
- Verification checklist

Optional Fields

- Statistical methods notes | OPTIONAL

- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Thresholds must be explicit; no “alert if bad.”
- Signals must map to alert routes and actions.
- Monitoring must avoid high-cardinality metrics that explode cost.

Output Format

1) Applicability

- applies: {{monitor.applies}} (true/false)
- 00_NA (if not applies): {{monitor.na_block}} | OPTIONAL

2) Monitoring Signals (canonical)

signal_id	target	type	metric	baseline	warn	critical	window	freq	route	action	notes
sig_01	{{signals[0].target}}	{{signals[0].type}}	{{signals[0].metric}}	{{signals[0].baseline}}	{{signals[0].warn}}	{{signals[0].critical}}	{{signals[0].window}}	{{signals[0].freq}}	{{signals[0].route}}	{{signals[0].action}}	{{signals[0].notes}}
sig_02	{{signals[1].target}}	{{signals[1].type}}	{{signals[1].metric}}	{{signals[1].baseline}}	{{signals[1].warn}}	{{signals[1].critical}}	{{signals[1].window}}	{{signals[1].freq}}	{{signals[1].route}}	{{signals[1].action}}	{{signals[1].notes}}

3) Escalation Process (required if applies)

- On warning: {{escalation.warning}}
- On critical: {{escalation.critical}}
- Backfill/pause authority: {{escalation.authority}} | OPTIONAL

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-02}} | OPTIONAL, {{xref:OBS-02}} | OPTIONAL
- Downstream: {{xref:DQV-05}} | OPTIONAL, {{xref:PIPE-04}} | OPTIONAL, {{xref:IRP-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Signals + thresholds + routes.
- advanced: Required if applies. Add escalation authority and methods notes.

Unknown Handling

- UNKNOWN_ALLOWED: `statistical_methods_notes`, `notes`, `authority`, `baseline` (if to be established but must be planned)
- If applies == true and thresholds are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_signals_count >= 12
    - thresholds_present == true
    - routes_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DQV-05

DQV-04 — Consistency & Integrity Monitoring (drift, anomalies)

Header Block

- template_id: DQV-04
- title: Consistency & Integrity Monitoring (drift, anomalies)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-04_Consistency_Integrity_Monitoring.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DQV-02", "OBS-02", "ALRT-01"]
- inputs_required: ["DQV-02", "OBS-02", "ALRT-01", "RELIA-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how the system monitors data consistency and integrity over time: drift detection, anomaly detection, referential integrity monitoring, and alerting/escalation when data quality degrades.

Inputs Required

- DQV-02: {{xref:DQV-02}} | OPTIONAL
- OBS-02: {{xref:OBS-02}} | OPTIONAL
- ALRT-01: {{xref:ALRT-01}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Monitoring signals catalog (minimum 12 signals)
- For each signal:
    - signal_id
    - target entity/dataset
    - signal type (drift/anomaly/integrity/freshness)
    - metric definition
    - baseline expectation
    - thresholds (warning/critical)
    - window
    - detection frequency
    - alert routing (who gets paged/ticket)
    - failure action (investigate/backfill/pause pipeline)
- Escalation process
- Verification checklist

Optional Fields

- Statistical methods notes | OPTIONAL

- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Thresholds must be explicit; no “alert if bad.”
- Signals must map to alert routes and actions.
- Monitoring must avoid high-cardinality metrics that explode cost.

Output Format

1) Applicability

- applies: {{monitor.applies}} (true/false)
- 00_NA (if not applies): {{monitor.na_block}} | OPTIONAL

2) Monitoring Signals (canonical)

signal_id	target	type	metric	baseline	warn	critical	window	freq	route	action	notes
sig_01	{{signals[0].target}}	{{signals[0].type}}	{{signals[0].metric}}	{{signals[0].baseline}}	{{signals[0].warn}}	{{signals[0].critical}}	{{signals[0].window}}	{{signals[0].freq}}	{{signals[0].route}}	{{signals[0].action}}	{{signals[0].notes}}
sig_02	{{signals[1].target}}	{{signals[1].type}}	{{signals[1].metric}}	{{signals[1].baseline}}	{{signals[1].warn}}	{{signals[1].critical}}	{{signals[1].window}}	{{signals[1].freq}}	{{signals[1].route}}	{{signals[1].action}}	{{signals[1].notes}}

3) Escalation Process (required if applies)

- On warning: {{escalation.warning}}
- On critical: {{escalation.critical}}
- Backfill/pause authority: {{escalation.authority}} | OPTIONAL

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-02}} | OPTIONAL, {{xref:OBS-02}} | OPTIONAL
- Downstream: {{xref:DQV-05}} | OPTIONAL, {{xref:PIPE-04}} | OPTIONAL, {{xref:IRP-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Signals + thresholds + routes.
- advanced: Required if applies. Add escalation authority and methods notes.

Unknown Handling

- UNKNOWN_ALLOWED: `statistical_methods_notes`, `notes`, `authority`, `baseline` (if to be established but must be planned)
- If applies == true and thresholds are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_signals_count >= 12
    - thresholds_present == true
    - routes_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DQV-06

DQV-05 — Bad Data Handling (quarantine, repair, backfill)

Header Block

- template_id: DQV-05
- title: Bad Data Handling (quarantine, repair, backfill)
- type: data_quality_validation
- template_version: 1.0.0
- output_path: 10_app/data_quality/DQV-05_Bad_Data_Handling.md
- compliance_gate_id: TMP-05.PRIMARY.DQV
- upstream_dependencies: ["DQV-02", "DQV-04", "PIPE-04", "WFO-05"]
- inputs_required: ["DQV-02", "DQV-04", "PIPE-04", "WFO-05", "ERR-02", "OBS-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how the system handles bad data when detected: quarantine, repair workflows, backfills, reconciliation, and safe reprocessing. This prevents silent corruption and ensures deterministic remediation.

Inputs Required

- DQV-02: {{xref:DQV-02}} | OPTIONAL
- DQV-04: {{xref:DQV-04}} | OPTIONAL
- PIPE-04: {{xref:PIPE-04}} | OPTIONAL

- WFO-05: {{xref:WFO-05}} | OPTIONAL
- ERR-02: {{xref:ERR-02}} | OPTIONAL
- OBS-04: {{xref:OBS-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Bad data classification (types):
    - schema invalid
    - semantic invalid
    - referential broken
    - duplicate
    - stale/out-of-order
- Quarantine model:
    - what data is quarantined
    - where it is stored
    - required quarantine record fields
    - retention policy
- Repair workflow definitions (minimum 6):
    - repair_id
    - trigger (which check/signal)
    - remediation steps
    - approvals required (if any)

    - safe re-run policy (idempotency)
    - verification checks
    - audit/logging requirements
- Backfill policy:
    - how backfills are scoped
    - rate limiting/impact controls
    - reprocessing rules
- Escalation rules (who owns fixing)
- Verification checklist

Optional Fields

- Auto-repair rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Quarantine must preserve enough context to debug but respect privacy/redaction.
- Repairs and backfills must be idempotent and observable.
- Any repair that changes user-visible outcomes must be auditable.
- Backfills must be throttled and safe for production.

Output Format

1) Bad Data Types (required)

- {{types[0]}}
- {{types[1]}}
- {{types[2]}} | OPTIONAL

2) Quarantine Model (required)

- Stored where: {{quarantine.location}}
- Required fields: {{quarantine.required_fields}}
- Retention policy: {{quarantine.retention}}
- Redaction rules: {{quarantine.redaction}} | OPTIONAL

3) Repair Workflows (canonical, min 6)

repair_id	trigger	steps	approvals	idempotency_rule	verify	audit	notes
rep_01	{{repairs[0].trigger}}	{{repairs[0].steps}}	{{repairs[0].approvals}}	{{repairs[0].idem}}	{{repairs[0].verify}}	{{repairs[0].audit}}	{{repairs[0].notes}}
rep_02	{{repairs[1].trigger}}	{{repairs[1].steps}}	{{repairs[1].approvals}}	{{repairs[1].idem}}	{{repairs[1].verify}}	{{repairs[1].audit}}	{{repairs[1].notes}}

4) Backfill Policy (required)

- Scope rules: {{backfill.scope}}
- Throttling rules: {{backfill.throttle}}
- Reprocessing rules: {{backfill.reprocess}}
- Stop conditions: {{backfill.stop_conditions}} | OPTIONAL

5) Escalation Rules (required)

- Owner assignment: {{escalation.owner_assignment}}

- SLA for critical issues: {{escalation.sla}} | OPTIONAL
- When to page: {{escalation.paging}} | OPTIONAL

6) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-04}} | OPTIONAL, {{xref:WFO-05}} | OPTIONAL, {{xref:PIPE-04}} | OPTIONAL
- Downstream: {{xref:DQV-06}} | OPTIONAL, {{xref:RELIA-05}} | OPTIONAL, {{xref:IRP-*}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Quarantine model + backfill policy basics.
- intermediate: Required. Add repair workflow catalog and verification checks.
- advanced: Required. Add approval governance, stop conditions, and audit rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `auto_repair`, `notes`, `stop_conditions`, `sla`
- If quarantine location or required fields are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DQV
- Pass conditions:
  - required_fields_present == true
  - quarantine_model_present == true
  - repairs_count >= 6
  - backfill_policy_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Search & Indexing (SRCH)

Search & Indexing (SRCH)

- SRCH-01 Search Scope & Surfaces (what is searchable, where)
- SRCH-02 Query Model (filters, ranking signals, facets)
- SRCH-03 Index Update Strategy (sync/async, reindex)
- SRCH-04 Search Result Quality Rules (relevance, freshness, dedupe)
- SRCH-05 Search Abuse Controls (gaming, spam, limits)
- SRCH-06 Search Observability (metrics, logging, evaluation)

SRCH-01

SRCH-01 — Search Scope & Surfaces (what is searchable, where)

Header Block

- template_id: SRCH-01
- title: Search Scope & Surfaces (what is searchable, where)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-01_Search_Scope_Surfaces.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["PRD-04", "DISC-03", "IAN-01", "DGL-04"]
- inputs_required: ["PRD-04", "DISC-03", "IAN-01", "DGL-04", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define what content/entities are searchable and where search appears in the product (surfaces): global search, per-page search, admin search, filters, and discovery surfaces. This prevents inconsistent search scope and makes indexing requirements deterministic.

Inputs Required

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- DISC-03: {{xref:DISC-03}} | OPTIONAL
- IAN-01: {{xref:IAN-01}} | OPTIONAL

- DGL-04: {{xref:DGL-04}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Search surfaces list (minimum 5 if search exists; otherwise mark N/A)
- For each surface:
    - surface_id
    - platform (web/mobile/admin)
    - location (nav, page, modal)
    - purpose
    - searchable entity types
    - permissions requirements (who can see results)
    - UX constraints pointer (DES/CDX/A11Y)
- Searchable entity inventory:
    - entity_id
    - fields searchable
    - sensitivity constraints (PII)
    - access control rule pointer (DGL-04/PMAD)
- Exclusions list (what must NOT be searchable)
- Coverage check: every searchable entity has an index strategy (SRCH-03)

Optional Fields

- SEO/public discovery notes | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Search results must respect authorization; never leak existence of private entities.
- Sensitive fields must be excluded or transformed (hash/redact).
- Surfaces must specify empty/error states pointers.

Output Format

1) Applicability

- applies: {{search.applies}} (true/false)
- 00_NA (if not applies): {{search.na_block}} | OPTIONAL

2) Search Surfaces (canonical)

surface_id	platform	location	purpose	entity_types	perms_rule_ref	ux_pointer	notes
srf_global	{{surfaces[0].platform}}	{{surfaces[0].location}}	{{surfaces[0].purpose}}	{{surfaces[0].entities}}	{{surfaces[0].perms}}	{{surfaces[0].ux}}	{{surfaces[0].notes}}
srf_admin	{{surfaces[1].platform}}	{{surfaces[1].location}}	{{surfaces[1].purpose}}	{{surfaces[1].entities}}	{{surfaces[1].perms}}	{{surfaces[1].ux}}	{{surfaces[1].notes}}

3) Searchable Entities (required if applies)

entity_id	searchable_fields	sensitivity	access_rule_ref	exclusions	notes

{{entities[0].id}}	{{entities[0].fields}}	{{entities[0].sens}}	{{entities[0].access}}	{{entities[0].exclusions}}	{{entities[0].notes}}

4) Exclusions (required if applies)

- Must NOT be searchable: {{exclusions.list}}
- Rationale: {{exclusions.rationale}} | OPTIONAL

5) Coverage Checks (required if applies)

- Every searchable entity has SRCH-03 plan: {{coverage.index_strategy_present}}
- Permissions defined for each surface: {{coverage.perms_present}}

Cross-References

- Upstream: {{xref:DISC-03}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL, {{xref:IAN-01}} | OPTIONAL
- Downstream: {{xref:SRCH-02}}, {{xref:SRCH-03}} | OPTIONAL, {{xref:SRCH-05}} | OPTIONAL, {{xref:SRCH-06}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required if applies. Surfaces + searchable entities list.
- intermediate: Required if applies. Add access rules and exclusions.
- advanced: Required if applies. Add coverage checks and sensitivity constraints.

Unknown Handling

- UNKNOWN_ALLOWED: `seo_notes`, `notes`, `ux_pointer` (if not designed yet but must be planned)
- If applies == true and access_rule_ref is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_surfaces_present == true
    - if_applies_then_entities_present == true
    - perms_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

SRCH-02

SRCH-01 — Search Scope & Surfaces (what is searchable, where)

Header Block

- template_id: SRCH-01
- title: Search Scope & Surfaces (what is searchable, where)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-01_Search_Scope_Surfaces.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["PRD-04", "DISC-03", "IAN-01", "DGL-04"]
- inputs_required: ["PRD-04", "DISC-03", "IAN-01", "DGL-04", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define what content/entities are searchable and where search appears in the product (surfaces): global search, per-page search, admin search, filters, and discovery surfaces. This prevents inconsistent search scope and makes indexing requirements deterministic.

Inputs Required

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- DISC-03: {{xref:DISC-03}} | OPTIONAL
- IAN-01: {{xref:IAN-01}} | OPTIONAL

- DGL-04: {{xref:DGL-04}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Search surfaces list (minimum 5 if search exists; otherwise mark N/A)
- For each surface:
    - surface_id
    - platform (web/mobile/admin)
    - location (nav, page, modal)
    - purpose
    - searchable entity types
    - permissions requirements (who can see results)
    - UX constraints pointer (DES/CDX/A11Y)
- Searchable entity inventory:
    - entity_id
    - fields searchable
    - sensitivity constraints (PII)
    - access control rule pointer (DGL-04/PMAD)
- Exclusions list (what must NOT be searchable)
- Coverage check: every searchable entity has an index strategy (SRCH-03)

Optional Fields

- SEO/public discovery notes | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Search results must respect authorization; never leak existence of private entities.
- Sensitive fields must be excluded or transformed (hash/redact).
- Surfaces must specify empty/error states pointers.

Output Format

1) Applicability

- applies: {{search.applies}} (true/false)
- 00_NA (if not applies): {{search.na_block}} | OPTIONAL

2) Search Surfaces (canonical)

surface_id	platform	location	purpose	entity_types	perms_rule_ref	ux_pointer	notes
srf_global	{{surfaces[0].platform}}	{{surfaces[0].location}}	{{surfaces[0].purpose}}	{{surfaces[0].entities}}	{{surfaces[0].perms}}	{{surfaces[0].ux}}	{{surfaces[0].notes}}
srf_admin	{{surfaces[1].platform}}	{{surfaces[1].location}}	{{surfaces[1].purpose}}	{{surfaces[1].entities}}	{{surfaces[1].perms}}	{{surfaces[1].ux}}	{{surfaces[1].notes}}

3) Searchable Entities (required if applies)

entity_id	searchable_fields	sensitivity	access_rule_ref	exclusions	notes

{{entities[0].id}}	{{entities[0].fields}}	{{entities[0].sens}}	{{entities[0].access}}	{{entities[0].exclusions}}	{{entities[0].notes}}

4) Exclusions (required if applies)

- Must NOT be searchable: {{exclusions.list}}
- Rationale: {{exclusions.rationale}} | OPTIONAL

5) Coverage Checks (required if applies)

- Every searchable entity has SRCH-03 plan: {{coverage.index_strategy_present}}
- Permissions defined for each surface: {{coverage.perms_present}}

Cross-References

- Upstream: {{xref:DISC-03}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL, {{xref:IAN-01}} | OPTIONAL
- Downstream: {{xref:SRCH-02}}, {{xref:SRCH-03}} | OPTIONAL, {{xref:SRCH-05}} | OPTIONAL, {{xref:SRCH-06}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required if applies. Surfaces + searchable entities list.
- intermediate: Required if applies. Add access rules and exclusions.
- advanced: Required if applies. Add coverage checks and sensitivity constraints.

Unknown Handling

- UNKNOWN_ALLOWED: `seo_notes`, `notes`, `ux_pointer` (if not designed yet but must be planned)
- If applies == true and access_rule_ref is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH
- Pass conditions:
    - required_fields_present == true
    - if_applies_then_surfaces_present == true
    - if_applies_then_entities_present == true
    - perms_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

SRCH-03

SRCH-03 — Index Update Strategy (sync/async, reindex)

Header Block

- template_id: SRCH-03
- title: Index Update Strategy (sync/async, reindex)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-03_Index_Update_Strategy.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["SRCH-01", "DGL-02", "WFO-01"]
- inputs_required: ["SRCH-01", "DGL-02", "WFO-01", "EVT-01", "ERR-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how search indexes are updated and kept consistent: synchronous vs asynchronous updates, event-driven indexing, reindex procedures, failure handling, and verification. This prevents stale search, index drift, and undefined reindex behavior.

Inputs Required

- SRCH-01: {{xref:SRCH-01}} | OPTIONAL
- DGL-02: {{xref:DGL-02}} | OPTIONAL
- WFO-01: {{xref:WFO-01}} | OPTIONAL

- EVT-01: {{xref:EVT-01}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Index inventory (minimum 3 indexes if search exists)
- For each index:
    - index_id
    - entity types covered
    - update mode (sync/async/hybrid)
    - triggers (write events, scheduled, manual)
    - pipeline steps (extract/transform/index)
    - freshness target (seconds/minutes)
    - failure posture (retry/DLQ/backfill)
    - idempotency/dedupe rule
    - permissions enforcement rule (no private leaks)
    - observability signals (lag, failure rate)
- Reindex procedure (high level):
    - when needed
    - how to run
    - how to verify
    - rollback posture

- Verification checklist

Optional Fields

- Blue/green index swap strategy | OPTIONAL
- Notes | OPTIONAL

Rules

- Any async indexing must specify freshness targets and lag monitoring.
- Updates must be idempotent and deduped.
- Permissions must be enforced at index time or query time (explicit choice).
- Reindex must be safe and verifiable; no “rebuild it and hope.”

Output Format

1) Index Inventory (canonical)

index_id	entities	update_mode	triggers	freshness_target	failure_posture	idem_dedupe	perms_rule	obs_signals	notes
idx_search_01	{{indexes[0].entities}}	{{indexes[0].mode}}	{{indexes[0].triggers}}	{{indexes[0].freshness}}	{{indexes[0].failure}}	{{indexes[0].idem}}	{{indexes[0].perms}}	{{indexes[0].obs}}	{{indexes[0].notes}}
idx_search_02	{{indexes[1].entities}}	{{indexes[1].mode}}	{{indexes[1].triggers}}	{{indexes[1].freshness}}	{{indexes[1].failure}}	{{indexes[1].idem}}	{{indexes[1].perms}}	{{indexes[1].obs}}	{{indexes[1].notes}}

2) Reindex Procedure (required)

- When needed: {{reindex.when}}

- How to run: {{reindex.how}}
- Verification: {{reindex.verify}}
- Rollback posture: {{reindex.rollback}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DGL-02}} | OPTIONAL, {{xref:WFO-01}} | OPTIONAL, {{xref:ERR-05}} | OPTIONAL
- Downstream: {{xref:SRCH-04}}, {{xref:SRCH-06}} | OPTIONAL, {{xref:OBS-04}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Index list + update modes + triggers.
- intermediate: Required. Add freshness targets and failure posture.
- advanced: Required. Add reindex verification and permissions enforcement approach.

Unknown Handling

- UNKNOWN_ALLOWED: `blue_green_swap`, `notes`, `rollback_posture`

- If any index lacks update_mode or triggers → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH
- Pass conditions:
    - required_fields_present == true
    - indexes_present == true
    - update_modes_defined == true
    - failure_posture_defined == true
    - reindex_procedure_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

SRCH-04

SRCH-04 — Search Result Quality Rules (relevance, freshness, dedupe)

Header Block

- template_id: SRCH-04
- title: Search Result Quality Rules (relevance, freshness, dedupe)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-04_Search_Result_Quality_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["SRCH-02", "SRCH-03", "DISC-04"]
- inputs_required: ["SRCH-02", "SRCH-03", "DISC-04", "PERF-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the quality rules and guardrails for search results: relevance expectations, freshness constraints, deduplication, pagination stability, and minimum quality thresholds. This makes search quality measurable and prevents regressions.

Inputs Required

- SRCH-02: {{xref:SRCH-02}} | OPTIONAL
- SRCH-03: {{xref:SRCH-03}} | OPTIONAL
- DISC-04: {{xref:DISC-04}} | OPTIONAL

- PERF-01: {{xref:PERF-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Quality dimensions:
    - relevance
    - freshness
    - dedupe
    - stability (no flicker)
- Rules catalog (minimum 15 rules)
- For each rule:
    - qrule_id
    - dimension
    - statement
    - metric/measurement method
    - threshold/target
    - failure action (alert/block release)
    - owner
- Dedupe policy (what counts as duplicate)
- Freshness policy per surface (time windows)
- Verification checklist

Optional Fields

- Human evaluation rubric | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Every rule must be measurable with a metric or eval method.
- Freshness rules must align with index update targets (SRCH-03).
- Dedupe must be deterministic; define tie-breaker order.

Output Format

1) Applicability

- applies: {{quality.applies}} (true/false)
- 00_NA (if not applies): {{quality.na_block}} | OPTIONAL

2) Quality Rules Catalog (canonical)

qrule_id	dimension	statement	measurement	threshold	failure_action	owner	notes
q_01	{{rules[0].dim}}	{{rules[0].stmt}}	{{rules[0].measure}}	{{rules[0].threshold}}	{{rules[0].action}}	{{rules[0].owner}}	{{rules[0].notes}}
q_02	{{rules[1].dim}}	{{rules[1].stmt}}	{{rules[1].measure}}	{{rules[1].threshold}}	{{rules[1].action}}	{{rules[1].owner}}	{{rules[1].notes}}

3) Dedupe Policy (required if applies)

- Duplicate definition: {{dedupe.definition}}

- Tie-breaker order: {{dedupe.tiebreaker}}
- Scope (within page/within query): {{dedupe.scope}} | OPTIONAL

4) Freshness Policy (required if applies)

surface_id	max_staleness	notes
{{freshness[0].surface}}	{{freshness[0].max_staleness}}	{{freshness[0].notes}}

5) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:SRCH-03}} | OPTIONAL, {{xref:DISC-04}} | OPTIONAL
- Downstream: {{xref:SRCH-06}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL, {{xref:ALRT-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Rules + dedupe + freshness per surface.
- advanced: Required if applies. Add measurement methods and failure actions rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `human_eval_rubric`, `notes`, `freshness_notes`
- If applies == true and thresholds are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH
- Pass conditions:
  - required_fields_present == true
  - if_applies_then_rules_count >= 15
  - measurement_defined == true
  - thresholds_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

SRCH-05

SRCH-05 — Search Abuse Controls (gaming, spam, limits)

Header Block

- template_id: SRCH-05
- title: Search Abuse Controls (gaming, spam, limits)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-05_Search_Abuse_Controls.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["SRCH-02", "TNS-01", "RLIM-01"]
- inputs_required: ["SRCH-02", "TNS-01", "RLIM-01", "DISC-05", "OBS-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how search is protected against abuse: query spam, scraping, ranking manipulation, keyword stuffing, and other gaming. This covers rate limits, detection heuristics, enforcement actions, and moderation hooks.

Inputs Required

- SRCH-02: {{xref:SRCH-02}} | OPTIONAL
- TNS-01: {{xref:TNS-01}} | OPTIONAL
- RLIM-01: {{xref:RLIM-01}} | OPTIONAL

- DISC-05: {{xref:DISC-05}} | OPTIONAL
- OBS-04: {{xref:OBS-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Abuse threat list (minimum 10 threats)
- Controls catalog (minimum 12 controls)
- For each control:
    - ctrl_id
    - threat(s) mitigated
    - layer (client/api/search service/index-time)
    - detection signal (metric/log pattern)
    - threshold
    - enforcement action (throttle/captcha/block/derank/quarantine)
    - appeal/recovery rule pointer
    - observability signals
- Rate limit policy pointer (RLIM)
- Ranking manipulation defenses:
    - dedupe/near-dup
    - content quality filters
    - anti-keyword stuffing rules
- Verification checklist

Optional Fields

- ML scoring pointer | OPTIONAL
- Notes | OPTIONAL

Rules

- Controls must be measurable and enforceable.
- Enforcement actions must be auditable and reversible where appropriate.
- Do not rely on client-only controls.
- Rate limits must be scoped (per user/ip/tenant).

Output Format

1) Threats (required)

- {{threats[0]}}
- {{threats[1]}}
- {{threats[2]}} | OPTIONAL

2) Controls Catalog (canonical)

ctrl_id	threats	layer	detection_signal	threshold	action	appeal_ptr	obs	notes
s_ctrl_01	{{controls[0].threats}}	{{controls[0].layer}}	{{controls[0].signal}}	{{controls[0].threshold}}	{{controls[0].action}}	{{controls[0].appeal}}	{{controls[0].obs}}	{{controls[0].notes}}
s_ctrl	{{controls[1].threats}}	{{controls[1].layer}}	{{controls[1].signal}}	{{controls[1].threshold}}	{{controls[1].action}}	{{controls[1].appeal}}	{{controls[1].obs}}	{{controls[1].notes}}

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3) Ranking Manipulation Defenses (required)

- Dedupe/near-dup rule: {{defenses.dedupe}}
- Content quality filters: {{defenses.quality_filters}}
- Keyword stuffing rules: {{defenses.keyword_stuffing}} | OPTIONAL

4) Rate Limit Pointer (required)

- RLIM pointer: {{xref:RLIM-01}} | OPTIONAL

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:TNS-01}} | OPTIONAL, {{xref:RLIM-01}} | OPTIONAL, {{xref:DISC-05}} | OPTIONAL
- Downstream: {{xref:ALRT-*}} | OPTIONAL, {{xref:SRCH-06}} | OPTIONAL
- Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Threat list + basic controls + rate-limit pointer.

- intermediate: Required. Add detection signals and thresholds.
- advanced: Required. Add enforcement/appeal rules and observability rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `ml_scoring`, `notes`, `keyword_stuffing_details`
- If any control lacks threshold or action → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH
- Pass conditions:
    - required_fields_present == true
    - threats_count >= 10
    - controls_count >= 12
    - thresholds_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

SRCH-06

SRCH-06 — Search Observability (metrics, logging, evaluation)

Header Block

- template_id: SRCH-06
- title: Search Observability (metrics, logging, evaluation)
- type: search_indexing
- template_version: 1.0.0
- output_path: 10_app/search/SRCH-06_Search_Observability.md
- compliance_gate_id: TMP-05.PRIMARY.SRCH
- upstream_dependencies: ["SRCH-03", "SRCH-04", "OBS-01"]
- inputs_required: ["SRCH-03", "SRCH-04", "OBS-01", "OBS-02", "OBS-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the required observability and evaluation signals for search: query metrics, latency, index lag, relevance evaluation, logging fields (redacted), dashboards, and alerting thresholds.

Inputs Required

- SRCH-03: {{xref:SRCH-03}} | OPTIONAL
- SRCH-04: {{xref:SRCH-04}} | OPTIONAL
- OBS-01: {{xref:OBS-01}} | OPTIONAL

- OBS-02: {{xref:OBS-02}} | OPTIONAL
- OBS-05: {{xref:OBS-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Metrics catalog (minimum 15 metrics):
    - query_count
    - p50/p95 latency
    - zero_results_rate
    - clickthrough/engagement (if applicable)
    - index_lag
    - reindex_failures
    - dedupe_rate
    - abuse blocks
- Logging field schema:
    - query_id
    - surface_id
    - normalized query (redacted policy)
    - filters applied
    - result_count
    - top_k ids (hashed or limited)

    - correlation_id
- Evaluation plan:
    - offline eval dataset
    - metrics (NDCG/MRR) if used
    - human review cadence (optional)
- Alert thresholds
- Verification checklist

Optional Fields

- Experimentation hooks pointer | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Do not log raw PII queries; define redaction/hashing.
- Metrics tags must avoid high-cardinality identifiers.
- Alerts must be tied to thresholds and routed.

Output Format

1) Applicability

- applies: {{search_obs.applies}} (true/false)
- 00_NA (if not applies): {{search_obs.na_block}} | OPTIONAL

2) Metrics Catalog (canonical)

metric	type	definition	tags_allowed	target	notes
search_queries_total	counter	{{metrics.q_total.def}}	{{metrics.q_total.tags}}	{{metrics.q_total.target}}	{{metrics.q_total.notes}}
search_latency_ms_p95	gauge/hist	{{metrics.lat_p95.def}}	{{metrics.lat_p95.tags}}	{{metrics.lat_p95.target}}	{{metrics.lat_p95.notes}}
search_zero_results_rate	rate	{{metrics.zero.def}}	{{metrics.zero.tags}}	{{metrics.zero.target}}	{{metrics.zero.notes}}

3) Logging Schema (required if applies)

field	required	redaction	notes
query_id	true	none	{{logs.query_id}}
surface_id	true	none	{{logs.surface_id}}
normalized_query	true	{{logs.redaction.query}}	{{logs.normalized_query_notes}}
filters	true	none	{{logs.filters}}
result_count	true	none	{{logs.result_count}}
correlation_id	true	none	{{logs.correlation_id}}

4) Evaluation Plan (required if applies)

- Offline dataset: {{eval.dataset}}
- Metrics used: {{eval.metrics}}
- Human review cadence: {{eval.human_cadence}} | OPTIONAL

5) Alert Thresholds (required if applies)

alert_id	metric	threshold	window	severity	route
s_alert_01	{{alerts[0].metric}}	{{alerts[0].threshold}}	{{alerts[0].window}}	{{alerts[0].severity}}	{{alerts[0].route}}

6) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:SRCH-04}} | OPTIONAL, {{xref:OBS-01}} | OPTIONAL
- Downstream: {{xref:ALRT-*}}* | *OPTIONAL, {{xref:EXPER-*}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Metrics + logging schema + alerts.
- advanced: Required if applies. Add evaluation plan rigor and dashboards pointers.

Unknown Handling

- UNKNOWN_ALLOWED: `experimentation_hooks`, `notes`, `human_review_cadence`
- If applies == true and logging redaction policy is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SRCH

- Pass conditions:
    - required_fields_present == true
    - if_applies_then_metrics_count >= 15
    - logging_schema_present == true
    - alert_thresholds_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Caching & Data Access Patterns (CACHE)

Caching & Data Access Patterns (CACHE)

- CACHE-01 Caching Strategy (what to cache, where)
- CACHE-02 Invalidation Rules (events, TTLs, busting)
- CACHE-03 Consistency Model (strong/eventual/stale-while-revalidate)
- CACHE-04 Read/Write Split Rules (read replicas, CQRS if used)
- CACHE-05 Rate/Cost Controls for Reads (hot keys, batching)
- CACHE-06 Cache Failure Behavior (fallbacks, degradation)

CACHE-01

CACHE-01 — Caching Strategy (what to cache, where)

Header Block

- template_id: CACHE-01
- title: Caching Strategy (what to cache, where)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-01_Caching_Strategy.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["DATA-07", "SRCH-01", "PERF-01"]
- inputs_required: ["DATA-07", "SRCH-01", "PERF-01", "COST-01", "CACHE-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the caching strategy: what is cached (data and responses), where caches live (client/server/CDN/edge/redis), why caching is needed, and what the allowed cache patterns are.

Inputs Required

- DATA-07: {{xref:DATA-07}} | OPTIONAL
- SRCH-01: {{xref:SRCH-01}} | OPTIONAL
- PERF-01: {{xref:PERF-01}} | OPTIONAL

- COST-01: {{xref:COST-01}} | OPTIONAL
- CACHE-03: {{xref:CACHE-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Cache layers inventory (client/server/edge/CDN)
- Cache candidates catalog (minimum 12)
- For each candidate:
    - cache_id
    - data/resource cached (entity/read_model/endpoint)
    - cache layer(s)
    - key pattern (deterministic)
    - TTL (or SWR policy)
    - invalidation trigger pointer (CACHE-02)
    - sensitivity/PII constraints
    - consistency requirement (strong/eventual)
    - expected benefit (latency/cost)
    - failure fallback behavior
- “Do not cache” list (sensitive or volatile)
- Verification checklist

Optional Fields

- Cache warming strategy | OPTIONAL
- Notes | OPTIONAL

Rules

- Cache keys must be deterministic and scoped (tenant/user where needed).
- Do not cache sensitive data without explicit policy and encryption/segmentation.
- Every cache entry must have TTL or invalidation; never “forever cache.”
- Cache candidates must map to data access patterns (DATA-08) and consistency model (CACHE-03).

Output Format

1) Cache Layers (required)

layer	location	typical_use	notes
client	{{layers.client.location}}	{{layers.client.use}}	{{layers.client.notes}}
server	{{layers.server.location}}	{{layers.server.use}}	{{layers.server.notes}}
edge	{{layers.edge.location}}	{{layers.edge.use}}	{{layers.edge.notes}}

2) Cache Candidates (canonical)

cache_id	cached_target	layer(s)	key_pattern	ttl_or_swr	invalidation_ref	sensitivity	consistency	benefit	fallback	notes
c_	{{candidates[	{{candidates[	{{candidate	{{candidat	{{candidates[0].i	{{candidate	{{candidates[0].c	{{candidates[	{{candidates[0	{{candidates[

01	0].target}}	0].layers}}	s[0].key}}	es[0].ttl}}	nvalidate}}	s[0].sens}}	onsistency}}	0].benefit}}	].fallback}}	0].notes}}
c_02	{{candidates[1].target}}	{{candidates[1].layers}}	{{candidates[1].key}}	{{candidates[1].ttl}}	{{candidates[1].invalidate}}	{{candidates[1].sens}}	{{candidates[1].consistency}}	{{candidates[1].benefit}}	{{candidates[1].fallback}}	{{candidates[1].notes}}

3) Do Not Cache (required)

- Do not cache list: {{no_cache.list}}
- Rationale: {{no_cache.rationale}} | OPTIONAL

4) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:CACHE-03}} | OPTIONAL, {{xref:DATA-07}} | OPTIONAL
- Downstream: {{xref:CACHE-02}}, {{xref:CACHE-06}} | OPTIONAL, {{xref:OBS-02}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Layers + candidate list + do-not-cache.
- intermediate: Required. Add key patterns, TTLs, and invalidation refs.
- advanced: Required. Add sensitivity constraints and fallback behavior.

Unknown Handling

- UNKNOWN_ALLOWED: `warming_strategy`, `notes`
- If any candidate lacks TTL/SWR and invalidation_ref → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE
- Pass conditions:
    - required_fields_present == true
    - candidates_count >= 12
    - keys_present == true
    - ttl_or_invalidation_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CACHE-02

CACHE-02 — Invalidation Rules (events, TTLs, busting)

Header Block

- template_id: CACHE-02
- title: Invalidation Rules (events, TTLs, busting)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-02_Invalidation_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["CACHE-01", "EVT-01", "WFO-01"]
- inputs_required: ["CACHE-01", "EVT-01", "WFO-01", "ERR-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic cache invalidation: what events or conditions invalidate what keys, TTL policies, busting strategies, and safe fallback behavior. This prevents stale data bugs and inconsistent cache behavior across layers.

Inputs Required

- CACHE-01: {{xref:CACHE-01}} | OPTIONAL
- EVT-01: {{xref:EVT-01}} | OPTIONAL
- WFO-01: {{xref:WFO-01}} | OPTIONAL

- ERR-05: {{xref:ERR-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Invalidation policy (global)
- Invalidation rule catalog (minimum 15 rules)
- For each rule:
    - inv_id
    - cache_id (from CACHE-01)
    - trigger (event/entity mutation/time/explicit bust)
    - keys affected (pattern)
    - propagation scope (client/server/edge)
    - execution mode (sync/async)
    - delay tolerance (max staleness)
    - failure handling (retry/skip/mark stale)
    - observability signals (invalidation success, lag)
- “Hard bust” policy (emergency invalidation)
- Verification checklist

Optional Fields

- Cache stampede mitigation rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Every cache candidate must have at least one invalidation rule OR explicit TTL-only policy.
- Invalidation triggers must map to real mutations/events (EVT) or workflow steps (WFO).
- Propagation must be defined across layers; no “invalidate somewhere.”
- Failure handling must avoid stale-for-ever.

Output Format

1) Global Policy (required)

- Default invalidation stance: {{policy.default}}
- TTL-only allowed when: {{policy.ttl_only_allowed}}
- Max staleness default: {{policy.max_staleness_default}} | OPTIONAL

2) Invalidation Rules (canonical)

inv_id	cache_id	trigger	keys_affected	scope	mode	max_staleness	failure_handling	obs	notes
inv_01	{{rules[0].cache_id}}	{{rules[0].trigger}}	{{rules[0].keys}}	{{rules[0].scope}}	{{rules[0].mode}}	{{rules[0].staleness}}	{{rules[0].failure}}	{{rules[0].obs}}	{{rules[0].notes}}
inv_02	{{rules[1].cache_id}}	{{rules[1].trigger}}	{{rules[1].keys}}	{{rules[1].scope}}	{{rules[1].mode}}	{{rules[1].staleness}}	{{rules[1].failure}}	{{rules[1].obs}}	{{rules[1].notes}}

3) Hard Bust Policy (required)

- When allowed: {{hard_bust.when}}
- Who can do it: {{hard_bust.who}}
- Required audit event: {{hard_bust.audit_event}} | OPTIONAL
- Safety checks: {{hard_bust.safety_checks}} | OPTIONAL

4) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:CACHE-01}} | OPTIONAL, {{xref:EVT-01}} | OPTIONAL, {{xref:WFO-01}} | OPTIONAL
- Downstream: {{xref:CACHE-06}} | OPTIONAL, {{xref:OBS-04}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global policy + rule catalog structure.
- intermediate: Required. Add propagation scope and max staleness.
- advanced: Required. Add failure handling and hard-bust governance.

Unknown Handling

- UNKNOWN_ALLOWED: `stampede_mitigation`, `notes`, `audit_event` (if not wired yet but must be planned)
- If any cache_id has no invalidation rule and no TTL-only justification → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE
- Pass conditions:
  - required_fields_present == true
  - rules_count >= 15
  - every_cache_has_invalidation_or_ttl_only == true
  - hard_bust_policy_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

CACHE-03

CACHE-03 — Consistency Model (strong/eventual/stale-while-revalidate)

Header Block

- template_id: CACHE-03
- title: Consistency Model (strong/eventual/stale-while-revalidate)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-03_Consistency_Model.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["DATA-07", "DATA-08", "ERR-05", "DES-05"]
- inputs_required: ["DATA-07", "DATA-08", "ERR-05", "DES-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical consistency guarantees for cached and read-model data: which parts of the system must be strongly consistent, which can be eventually consistent, and where stale-while-revalidate (SWR) is allowed. This aligns backend behavior with UX expectations and prevents “mystery staleness.”

Inputs Required

- DATA-07: {{xref:DATA-07}} | OPTIONAL
- DATA-08: {{xref:DATA-08}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL

- DES-05: {{xref:DES-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Consistency modes definitions:
    - strong
    - eventual
    - stale-while-revalidate (SWR)
- Consistency requirements by domain surface (minimum 12 entries):
    - surface/operation (screen_id/endpoint_id/read_model_id/cache_id)
    - consistency mode
    - maximum staleness (if not strong)
    - user-visible behavior (loading/stale badge/retry)
    - reconciliation behavior (when fresh data arrives)
    - reason_code/UX mapping pointer (ERR/DES/CDX) | OPTIONAL
- Default consistency stance (system-wide)
- Exceptions policy (when strong is mandatory)

Optional Fields

- Multi-region notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Any operation that changes money, permissions, or critical state should default to strong consistency unless explicitly justified.
- SWR must specify UI behavior and max staleness.
- Eventual consistency must specify convergence path (what triggers revalidation).
- Consistency definitions must be used by CACHE-01 candidates and DATA-07 read models.

Output Format

1) Mode Definitions (required)

- strong: {{modes.strong}}
- eventual: {{modes.eventual}}
- swr: {{modes.swr}}

2) Consistency Requirements Matrix (canonical)

target	kind (screen/endpoint/read_model/cache)	mode	max_staleness	user_visible_behavior	reconcile_behavior	notes
{{matrix[0].target}}	{{matrix[0].kind}}	{{matrix[0].mode}}	{{matrix[0].staleness}}	{{matrix[0].ux}}	{{matrix[0].reconcile}}	{{matrix[0].notes}}
{{matrix[1].target}}	{{matrix[1].kind}}	{{matrix[1].mode}}	{{matrix[1].staleness}}	{{matrix[1].ux}}	{{matrix[1].reconcile}}	{{matrix[1].notes}}

3) Default Stance (required)

- Default mode: {{defaults.mode}}
- Default max staleness (if applicable): {{defaults.max_staleness}} | OPTIONAL

4) Exceptions Policy (required)

- Strong required when: {{exceptions.strong_required_when}}
- Approval needed for eventual/SWR on critical flows: {{exceptions.approval}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-07}} | OPTIONAL, {{xref:DES-05}} | OPTIONAL, {{xref:ERR-05}} | OPTIONAL
- Downstream: {{xref:CACHE-01}}, {{xref:CACHE-02}} | OPTIONAL, {{xref:CACHE-06}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Mode definitions + default stance.
- intermediate: Required. Add matrix with max staleness + UX behavior.
- advanced: Required. Add reconciliation rules and exceptions governance.

Unknown Handling

- UNKNOWN_ALLOWED: `multi_region_notes`, `notes`, `approval_policy`
- If any non-strong entry lacks max_staleness or user-visible behavior → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE

- Pass conditions:
    - required_fields_present == true
    - mode_definitions_present == true
    - matrix_present == true
    - non_strong_have_staleness_and_ux == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CACHE-04

CACHE-04 — Read/Write Split Rules (read replicas, CQRS if used)

Header Block

- template_id: CACHE-04
- title: Read/Write Split Rules (read replicas, CQRS if used)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-04_Read_Write_Split_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["DATA-07", "DATA-08", "SBDT-02"]
- inputs_required: ["DATA-07", "DATA-08", "SBDT-02", "PERF-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define when and how the system splits reads and writes: use of read replicas, CQRS patterns, read models, and consistency constraints. This ensures predictable behavior under replication lag and prevents incorrect reads after writes.

Inputs Required

- DATA-07: {{xref:DATA-07}} | OPTIONAL
- DATA-08: {{xref:DATA-08}} | OPTIONAL
- SBDT-02: {{xref:SBDT-02}} | OPTIONAL

- PERF-02: {{xref:PERF-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Read/write split stance (none/replicas/CQRS/hybrid)
- Rules for routing:
    - what must go to primary
    - what may go to replicas
    - “read-your-writes” guarantees (if any)
- Lag handling:
    - maximum tolerated replication lag
    - stale read behaviors
    - fallback to primary rules
- CQRS rules (if used):
    - command model vs query model boundaries
    - projection update mechanism pointer (WFO/SRCH/CACHE)
- Verification checklist

Optional Fields

- Multi-region replication notes | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- After write operations, critical reads must not use stale replicas unless explicitly allowed.
- Lag must be monitored; define alert thresholds.
- CQRS must specify projection freshness and failure posture.

Output Format

1) Applicability

- applies: {{rw.applies}} (true/false)
- 00_NA (if not applies): {{rw.na_block}} | OPTIONAL

2) Stance (required if applies)

- Stance: {{rw.stance}} (none/replicas/cqrs/hybrid)
- Why: {{rw.why}}

3) Routing Rules (required if applies)

- Must use primary for: {{routing.primary_required}}
- May use replicas for: {{routing.replica_allowed}}
- Read-your-writes guarantee: {{routing.read_your_writes}} | OPTIONAL

4) Lag Handling (required if applies)

- Max tolerated lag: {{lag.max}}
- Behavior when lag exceeded: {{lag.behavior}}

- Fallback rule: {{lag.fallback}} | OPTIONAL

5) CQRS Rules (required if used)

- Boundaries: {{cqrs.boundaries}} | OPTIONAL
- Projection mechanism pointer: {{cqrs.projection_pointer}} | OPTIONAL
- Freshness target: {{cqrs.freshness}} | OPTIONAL
- Failure posture: {{cqrs.failure}} | OPTIONAL

6) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DATA-07}} | OPTIONAL, {{xref:SBDT-02}} | OPTIONAL
- Downstream: {{xref:CACHE-03}} | OPTIONAL, {{xref:PERF-03}} | OPTIONAL, {{xref:OBS-04}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required.
- intermediate: Required if applies. Stance + routing + lag policy.
- advanced: Required if applies. Add CQRS boundaries and monitoring thresholds.

Unknown Handling

- UNKNOWN_ALLOWED: `multi_region_notes`, `notes`, `read_your_writes`, `cqrs_rules`
- If applies == true and max tolerated lag is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE
- Pass conditions:
  - required_fields_present == true
  - if_applies_then_stance_present == true
  - if_applies_then_routing_rules_present == true
  - if_applies_then_lag_policy_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

CACHE-05

CACHE-05 — Rate/Cost Controls for Reads (hot keys, batching)

Header Block

- template_id: CACHE-05
- title: Rate/Cost Controls for Reads (hot keys, batching)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-05_Rate_Cost_Controls_for_Reads.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["CACHE-01", "PERF-02", "COST-01"]
- inputs_required: ["CACHE-01", "PERF-02", "COST-01", "RLIM-01", "OBS-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define controls that keep read load and cost bounded: batching, request coalescing, hot-key mitigation, per-tenant/user limits, cache tiering, and fail-open/close behaviors under high load.

Inputs Required

- CACHE-01: {{xref:CACHE-01}} | OPTIONAL
- PERF-02: {{xref:PERF-02}} | OPTIONAL
- COST-01: {{xref:COST-01}} | OPTIONAL

- RLIM-01: {{xref:RLIM-01}} | OPTIONAL
- OBS-02: {{xref:OBS-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Control catalog (minimum 12 controls)
- For each control:
    - ctrl_id
    - target layer (client/server/edge/db/cache)
    - mechanism (batching/coalescing/throttle/circuit/priority)
    - scope (per user/tenant/ip/global)
    - trigger metric (qps, p95 latency, cache miss rate)
    - threshold
    - action taken (reduce, shed load, degrade)
    - user impact (UX behavior pointer)
    - observability signals (metrics/alerts)
    - rollback/disable rule
- Hot key policy:
    - detection
    - mitigation (partitioning, local caches, jitter)
- Verification checklist

Optional Fields

- Tenant tiering policy | OPTIONAL
- Notes | OPTIONAL

Rules

- Controls must be measurable and reversible.
- User-impacting degradation must have defined UX behavior.
- Rate limiting must not break critical system functions; define allow-lists.
- Hot key mitigation must avoid creating new hotspots.

Output Format

1) Controls Catalog (canonical)

ctrl_id	layer	mechanism	scope	trigger_metric	threshold	action	user_impact	obs_signals	rollback_rule	notes
read_ctrl_01	{{controls[0].layer}}	{{controls[0].mech}}	{{controls[0].scope}}	{{controls[0].metric}}	{{controls[0].threshold}}	{{controls[0].action}}	{{controls[0].impact}}	{{controls[0].obs}}	{{controls[0].rollback}}	{{controls[0].notes}}
read_ctrl_02	{{controls[1].layer}}	{{controls[1].mech}}	{{controls[1].scope}}	{{controls[1].metric}}	{{controls[1].threshold}}	{{controls[1].action}}	{{controls[1].impact}}	{{controls[1].obs}}	{{controls[1].rollback}}	{{controls[1].notes}}

2) Hot Key Policy (required)

- Detection: {{hotkeys.detect}}
- Mitigation: {{hotkeys.mitigate}}

- Alert thresholds: {{hotkeys.alerts}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RLIM-01}} | OPTIONAL, {{xref:PERF-02}} | OPTIONAL, {{xref:CACHE-01}} | OPTIONAL
- Downstream: {{xref:CACHE-06}} | OPTIONAL, {{xref:ALRT-*}} | OPTIONAL, {{xref:PERF-05}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Basic controls + hot key detection.
- intermediate: Required. Add thresholds, actions, and rollback rules.
- advanced: Required. Add tiering and UX degradation behavior rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `tenant_tiering_policy`, `notes`, `allow_lists`, `alerts`
- If any control lacks trigger_metric or action → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE
- Pass conditions:
    - required_fields_present == true
    - controls_count >= 12
    - thresholds_present == true
    - rollback_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CACHE-06

CACHE-06 — Cache Failure Behavior (fallbacks, degradation)

Header Block

- template_id: CACHE-06
- title: Cache Failure Behavior (fallbacks, degradation)
- type: caching_data_access_patterns
- template_version: 1.0.0
- output_path: 10_app/caching/CACHE-06_Cache_Failure_Behavior.md
- compliance_gate_id: TMP-05.PRIMARY.CACHE
- upstream_dependencies: ["CACHE-01", "CACHE-02", "CACHE-03", "RELIA-01"]
- inputs_required: ["CACHE-01", "CACHE-02", "CACHE-03", "RELIA-01", "ERR-01", "DES-07", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define what happens when caches fail or misbehave: cache outages, high miss rates, stale data, stampedes, and partial invalidation failures. This specifies deterministic fallback paths and UX degradation behavior.

Inputs Required

- CACHE-01: {{xref:CACHE-01}} | OPTIONAL
- CACHE-02: {{xref:CACHE-02}} | OPTIONAL
- CACHE-03: {{xref:CACHE-03}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- ERR-01: {{xref:ERR-01}} | OPTIONAL
- DES-07: {{xref:DES-07}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Failure mode catalog (minimum 10)
- For each failure mode:
    - fail_id
    - cache layer affected (client/server/edge)
    - detection signal (timeout, error rate, miss spike)
    - expected behavior:
        - fail open vs fail closed
        - fallback source (DB/read model/stale copy)
        - max staleness allowed
    - user-visible behavior (loading, banner, stale badge)
    - retry/backoff rules (avoid stampedes)
    - circuit breaker rule (when to stop using cache)
    - observability and alerting (what triggers paging)
- Stampede mitigation policy
- Verification checklist

Optional Fields

- Emergency disable switch policy | OPTIONAL
- Notes | OPTIONAL

Rules

- Failure behaviors must align with CACHE-03 consistency requirements.
- Avoid cascading failures: do not hammer DB when cache is down without throttling.
- User-visible behavior must be defined for any user-facing impact.
- “Fail open” must not leak data; enforce auth in fallback path.

Output Format

1) Failure Modes (canonical)

fail_id	layer	detection	behavior	fallback_source	max_staleness	user_behavior	retry_backoff	circuit_rule	alerting	notes
cache_fail_01	{{fails[0].layer}}	{{fails[0].detect}}	{{fails[0].behavior}}	{{fails[0].fallback}}	{{fails[0].staleness}}	{{fails[0].ux}}	{{fails[0].retry}}	{{fails[0].circuit}}	{{fails[0].alert}}	{{fails[0].notes}}
cache_fail_02	{{fails[1].layer}}	{{fails[1].detect}}	{{fails[1].behavior}}	{{fails[1].fallback}}	{{fails[1].staleness}}	{{fails[1].ux}}	{{fails[1].retry}}	{{fails[1].circuit}}	{{fails[1].alert}}	{{fails[1].notes}}

2) Stampede Mitigation (required)

- Coalescing strategy: {{stampede.coalescing}}
- Jitter rules: {{stampede.jitter}} | OPTIONAL
- Max concurrent refreshes: {{stampede.max_refresh}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:CACHE-03}} | OPTIONAL, {{xref:RELIA-01}} | OPTIONAL, {{xref:DES-07}} | OPTIONAL
- Downstream: {{xref:ALRT-*}}* | *OPTIONAL, {{xref:IRP-}}* | OPTIONAL, {{xref:PERF-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Basic failure modes + fallback paths.
- intermediate: Required. Add detection signals, max staleness, and circuit rules.
- advanced: Required. Add stampede mitigation and alerting thresholds rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `emergency_disable_switch`, `notes`, `jitter_rules`, `max_refresh`
- If any failure mode lacks fallback_source or user_behavior (when user-facing) → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CACHE
- Pass conditions:
    - required_fields_present == true
    - failure_modes_count >= 10
    - fallback_paths_present == true
    - stampede_policy_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Reporting & Aggregations (RPT)

Reporting & Aggregations (RPT)

- RPT-01 Reporting Surfaces Inventory (dashboards, exports, admin)
- RPT-02 Metrics Definitions (canonical KPI semantics)
- RPT-03 Aggregation & Rollup Rules (windows, group-bys)
- RPT-04 Snapshotting Strategy (daily/weekly, recompute rules)
- RPT-05 Data Access & Permissions for Reports
- RPT-06 Reporting Data Quality Rules (reconciliation, correctness)

RPT-01

RPT-01 — Reporting Surfaces Inventory (dashboards, exports, admin)

Header Block

- template_id: RPT-01
- title: Reporting Surfaces Inventory (dashboards, exports, admin)
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-01_Reporting_Surfaces_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["SMIP-01", "RPT-02", "DGL-04"]
- inputs_required: ["SMIP-01", "RPT-02", "DGL-04", "ADMIN-01", "DIST-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define where reporting exists in the product: dashboards, admin views, exports, scheduled reports, and external BI surfaces. This makes reporting scope deterministic and ties it to permissions, distribution, and metric definitions.

Inputs Required

- SMIP-01: {{xref:SMIP-01}} | OPTIONAL
- RPT-02: {{xref:RPT-02}} | OPTIONAL
- DGL-04: {{xref:DGL-04}} | OPTIONAL

- ADMIN-01: {{xref:ADMIN-01}} | OPTIONAL
- DIST-03: {{xref:DIST-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Reporting surfaces list (minimum 6 if product has reporting; otherwise mark N/A)
- For each surface:
    - surface_id
    - platform (web/mobile/admin/external)
    - location (route/screen_id/export type)
    - purpose
    - audience (roles)
    - metrics included (metric_id list)
    - data freshness expectation (real-time/hourly/daily)
    - delivery mode (interactive/export/scheduled)
    - permissions enforcement rule pointer (DGL-04/PMAD)
    - export constraints (PII redaction, limits)
    - observability signals (usage, failures)
- Exclusions (what is NOT available)
- Coverage check: surfaces reference RPT-02 metrics definitions

Optional Fields

- Customer-facing vs internal separation notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Surfaces must not expose metrics without a canonical definition (RPT-02).
- Exports are privileged for sensitive data; define redaction and approvals.
- Freshness expectations must align with aggregation/snapshot strategy (RPT-03/RPT-04).
- Permissions must be explicit; “admin-only” still needs role definition.

Output Format

1) Applicability

- applies: {{reporting.applies}} (true/false)
- 00_NA (if not applies): {{reporting.na_block}} | OPTIONAL

2) Reporting Surfaces (canonical)

surface_id	platform	location	purpose	audience_roles	metrics	freshness	delivery	perms_ref	export_constraints	obs	notes
rpt_srf_01	{{surfaces[0].platform}}	{{surfaces[0].location}}	{{surfaces[0].purpose}}	{{surfaces[0].audience}}	{{surfaces[0].metrics}}	{{surfaces[0].freshness}}	{{surfaces[0].delivery}}	{{surfaces[0].perms}}	{{surfaces[0].export}}	{{surfaces[0].obs}}	{{surfaces[0].notes}}

rpt_srf_02	{{surfaces[1].platform}}	{{surfaces[1].location}}	{{surfaces[1].purpose}}	{{surfaces[1].audience}}	{{surfaces[1].metrics}}	{{surfaces[1].freshness}}	{{surfaces[1].delivery}}	{{surfaces[1].perms}}	{{surfaces[1].export}}	{{surfaces[1].obs}}	{{surfaces[1].notes}}

3) Exclusions (required if applies)

- Not available: {{exclusions.list}}
- Rationale: {{exclusions.rationale}} | OPTIONAL

4) Coverage Checks (required if applies)

- All surfaces reference RPT-02 metrics: {{coverage.metrics_defined}}
- All surfaces have permissions refs: {{coverage.perms_defined}}

Cross-References

- Upstream: {{xref:SMIP-01}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL
- Downstream: {{xref:RPT-03}}, {{xref:RPT-04}}, {{xref:RPT-05}} | OPTIONAL, {{xref:RPT-06}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

RPT-02

RPT-01 — Reporting Surfaces Inventory (dashboards, exports, admin)

Header Block

- template_id: RPT-01
- title: Reporting Surfaces Inventory (dashboards, exports, admin)
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-01_Reporting_Surfaces_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["SMIP-01", "RPT-02", "DGL-04"]
- inputs_required: ["SMIP-01", "RPT-02", "DGL-04", "ADMIN-01", "DIST-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define where reporting exists in the product: dashboards, admin views, exports, scheduled reports, and external BI surfaces. This makes reporting scope deterministic and ties it to permissions, distribution, and metric definitions.

Inputs Required

- SMIP-01: {{xref:SMIP-01}} | OPTIONAL
- RPT-02: {{xref:RPT-02}} | OPTIONAL
- DGL-04: {{xref:DGL-04}} | OPTIONAL

- ADMIN-01: {{xref:ADMIN-01}} | OPTIONAL
- DIST-03: {{xref:DIST-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Reporting surfaces list (minimum 6 if product has reporting; otherwise mark N/A)
- For each surface:
    - surface_id
    - platform (web/mobile/admin/external)
    - location (route/screen_id/export type)
    - purpose
    - audience (roles)
    - metrics included (metric_id list)
    - data freshness expectation (real-time/hourly/daily)
    - delivery mode (interactive/export/scheduled)
    - permissions enforcement rule pointer (DGL-04/PMAD)
    - export constraints (PII redaction, limits)
    - observability signals (usage, failures)
- Exclusions (what is NOT available)
- Coverage check: surfaces reference RPT-02 metrics definitions

Optional Fields

- Customer-facing vs internal separation notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Surfaces must not expose metrics without a canonical definition (RPT-02).
- Exports are privileged for sensitive data; define redaction and approvals.
- Freshness expectations must align with aggregation/snapshot strategy (RPT-03/RPT-04).
- Permissions must be explicit; “admin-only” still needs role definition.

Output Format

1) Applicability

- applies: {{reporting.applies}} (true/false)
- 00_NA (if not applies): {{reporting.na_block}} | OPTIONAL

2) Reporting Surfaces (canonical)

surface_id	platform	location	purpose	audience_roles	metrics	freshness	delivery	perms_ref	export_constraints	obs	notes
rpt_srf_01	{{surfaces[0].platform}}	{{surfaces[0].location}}	{{surfaces[0].purpose}}	{{surfaces[0].audience}}	{{surfaces[0].metrics}}	{{surfaces[0].freshness}}	{{surfaces[0].delivery}}	{{surfaces[0].perms}}	{{surfaces[0].export}}	{{surfaces[0].obs}}	{{surfaces[0].notes}}

rpt_srf_02	{{surfaces[1].platform}}	{{surfaces[1].location}}	{{surfaces[1].purpose}}	{{surfaces[1].audience}}	{{surfaces[1].metrics}}	{{surfaces[1].freshness}}	{{surfaces[1].delivery}}	{{surfaces[1].perms}}	{{surfaces[1].export}}	{{surfaces[1].obs}}	{{surfaces[1].notes}}

3) Exclusions (required if applies)

- Not available: {{exclusions.list}}
- Rationale: {{exclusions.rationale}} | OPTIONAL

4) Coverage Checks (required if applies)

- All surfaces reference RPT-02 metrics: {{coverage.metrics_defined}}
- All surfaces have permissions refs: {{coverage.perms_defined}}

Cross-References

- Upstream: {{xref:SMIP-01}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL
- Downstream: {{xref:RPT-03}}, {{xref:RPT-04}}, {{xref:RPT-05}} | OPTIONAL, {{xref:RPT-06}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

RPT-03

RPT-03 — Aggregation & Rollup Rules (windows, group-bys)

Header Block

- template_id: RPT-03
- title: Aggregation & Rollup Rules (windows, group-bys)
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-03_Aggregation_Rollup_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["RPT-02", "DATA-07", "BI-02"]
- inputs_required: ["RPT-02", "DATA-07", "BI-02", "DQV-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how metrics are aggregated and rolled up: time windows, grouping dimensions, late-arriving data handling, deduplication rules, and recomputation policies. This ensures report numbers are stable, explainable, and consistent across surfaces.

Inputs Required

- RPT-02: {{xref:RPT-02}} | OPTIONAL
- DATA-07: {{xref:DATA-07}} | OPTIONAL
- BI-02: {{xref:BI-02}} | OPTIONAL

- DQV-02: {{xref:DQV-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Rollup rules catalog (minimum 15 rules)
- For each rule:
    - rollup_id
    - metric_id(s)
    - window type (hour/day/week/month/rolling)
    - window boundaries (timezone)
    - group-by dimensions allowed
    - dedupe rule (event_id/entity_id)
    - late-arriving data policy (backfill window)
    - recompute policy (full/partial)
    - correctness checks (reconciliation rule)
    - owner
- Global rules:
    - timezone standard for windows
    - rounding rules
    - null/empty handling
- Verification checklist

Optional Fields

- Cross-metric consistency rules | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Every rollup must specify timezone and dedupe keys.
- Late-arriving data must have explicit backfill windows.
- Recompute policies must be deterministic and observable.

Output Format

1) Applicability

- applies: {{rollups.applies}} (true/false)
- 00_NA (if not applies): {{rollups.na_block}} | OPTIONAL

2) Rollup Rules Catalog (canonical)

rollup_id	metrics	window	timezone	group_by	dedupe_key	late_data_policy	recompute	checks	owner	notes
roll_01	{{rules[0].metrics}}	{{rules[0].window}}	{{rules[0].tz}}	{{rules[0].group_by}}	{{rules[0].dedupe}}	{{rules[0].late}}	{{rules[0].recompute}}	{{rules[0].checks}}	{{rules[0].owner}}	{{rules[0].notes}}
roll_02	{{rules[1].metrics}}	{{rules[1].window}}	{{rules[1].tz}}	{{rules[1].group_by}}	{{rules[1].dedupe}}	{{rules[1].late}}	{{rules[1].recompute}}	{{rules[1].checks}}	{{rules[1].owner}}	{{rules[1].notes}}

3) Global Rules (required if applies)

- Timezone standard: {{global.timezone}}
- Rounding rules: {{global.rounding}}
- Null/empty handling: {{global.null_handling}}

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RPT-02}} | OPTIONAL, {{xref:DQV-02}} | OPTIONAL
- Downstream: {{xref:RPT-04}}, {{xref:RPT-06}} | OPTIONAL, {{xref:BI-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

RPT-04

RPT-03 — Aggregation & Rollup Rules (windows, group-bys)

Header Block

- template_id: RPT-03
- title: Aggregation & Rollup Rules (windows, group-bys)
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-03_Aggregation_Rollup_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["RPT-02", "DATA-07", "BI-02"]
- inputs_required: ["RPT-02", "DATA-07", "BI-02", "DQV-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how metrics are aggregated and rolled up: time windows, grouping dimensions, late-arriving data handling, deduplication rules, and recomputation policies. This ensures report numbers are stable, explainable, and consistent across surfaces.

Inputs Required

- RPT-02: {{xref:RPT-02}} | OPTIONAL
- DATA-07: {{xref:DATA-07}} | OPTIONAL
- BI-02: {{xref:BI-02}} | OPTIONAL

- DQV-02: {{xref:DQV-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Rollup rules catalog (minimum 15 rules)
- For each rule:
    - rollup_id
    - metric_id(s)
    - window type (hour/day/week/month/rolling)
    - window boundaries (timezone)
    - group-by dimensions allowed
    - dedupe rule (event_id/entity_id)
    - late-arriving data policy (backfill window)
    - recompute policy (full/partial)
    - correctness checks (reconciliation rule)
    - owner
- Global rules:
    - timezone standard for windows
    - rounding rules
    - null/empty handling
- Verification checklist

Optional Fields

- Cross-metric consistency rules | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Every rollup must specify timezone and dedupe keys.
- Late-arriving data must have explicit backfill windows.
- Recompute policies must be deterministic and observable.

Output Format

1) Applicability

- applies: {{rollups.applies}} (true/false)
- 00_NA (if not applies): {{rollups.na_block}} | OPTIONAL

2) Rollup Rules Catalog (canonical)

rollup_id	metrics	window	timezone	group_by	dedupe_key	late_data_policy	recompute	checks	owner	notes
roll_01	{{rules[0].metrics}}	{{rules[0].window}}	{{rules[0].tz}}	{{rules[0].group_by}}	{{rules[0].dedupe}}	{{rules[0].late}}	{{rules[0].recompute}}	{{rules[0].checks}}	{{rules[0].owner}}	{{rules[0].notes}}
roll_02	{{rules[1].metrics}}	{{rules[1].window}}	{{rules[1].tz}}	{{rules[1].group_by}}	{{rules[1].dedupe}}	{{rules[1].late}}	{{rules[1].recompute}}	{{rules[1].checks}}	{{rules[1].owner}}	{{rules[1].notes}}

3) Global Rules (required if applies)

- Timezone standard: {{global.timezone}}
- Rounding rules: {{global.rounding}}
- Null/empty handling: {{global.null_handling}}

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RPT-02}} | OPTIONAL, {{xref:DQV-02}} | OPTIONAL
- Downstream: {{xref:RPT-04}}, {{xref:RPT-06}} | OPTIONAL, {{xref:BI-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

RPT-05

RPT-05 — Data Access & Permissions for Reports

Header Block

- template_id: RPT-05
- title: Data Access & Permissions for Reports
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-05_Data_Access_Permissions_for_Reports.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["RPT-01", "DGL-04", "PMAD-02"]
- inputs_required: ["RPT-01", "DGL-04", "PMAD-02", "DGP-01", "ADMIN-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define reporting-specific permissions: who can view which dashboards, slices, and exports; what row-level/tenant-level constraints apply; what redaction rules apply; and how report access is audited.

Inputs Required

- RPT-01: {{xref:RPT-01}} | OPTIONAL
- DGL-04: {{xref:DGL-04}} | OPTIONAL
- PMAD-02: {{xref:PMAD-02}} | OPTIONAL

- DGP-01: {{xref:DGP-01}} | OPTIONAL
- ADMIN-01: {{xref:ADMIN-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Report access matrix (minimum 20 entries)
- For each entry:
    - rpt_surface_id (from RPT-01)
    - metric_id(s) visible
    - audience roles
    - row-level constraints (tenant/org/owner)
    - export allowed (yes/no)
    - export redaction rules (if allowed)
    - approval required (yes/no + who)
    - audit event requirement
    - anti-reidentification rule (small cohorts)
- Global rules:
    - default deny for report access
    - minimum cohort size rules for breakdowns
    - export limits (rows/day, etc.)
- Verification checklist

Optional Fields

- External sharing rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Reporting access must align with PMAD/DGL access controls.
- Exports of sensitive data require stricter approvals and redaction.
- Small cohort breakdowns must be blocked or bucketed to prevent reidentification.
- Access to report exports should be auditable.

Output Format

1) Report Access Matrix (canonical)

rpt_surface_id	metrics	roles	row_constraints	export_allowed	export_redaction	approval	audit_event	cohort_rules	notes
{{matrix[0].surface}}	{{matrix[0].metrics}}	{{matrix[0].roles}}	{{matrix[0].rows}}	{{matrix[0].export}}	{{matrix[0].redaction}}	{{matrix[0].approval}}	{{matrix[0].audit}}	{{matrix[0].cohort}}	{{matrix[0].notes}}
{{matrix[1].surface}}	{{matrix[1].metrics}}	{{matrix[1].roles}}	{{matrix[1].rows}}	{{matrix[1].export}}	{{matrix[1].redaction}}	{{matrix[1].approval}}	{{matrix[1].audit}}	{{matrix[1].cohort}}	{{matrix[1].notes}}

2) Global Rules (required)

- Default deny: {{global.default_deny}}
- Minimum cohort size: {{global.min_cohort_size}}
- Export limits: {{global.export_limits}} | OPTIONAL
- Bucketing/anonymization rules: {{global.bucketing}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RPT-01}} | OPTIONAL, {{xref:DGL-04}} | OPTIONAL, {{xref:PMAD-02}} | OPTIONAL
- Downstream: {{xref:RPT-06}} | OPTIONAL, {{xref:COMP-02}} | OPTIONAL, {{xref:ADMIN-03}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Matrix + default deny + basic cohort rules.
- intermediate: Required. Add export constraints and audit events.
- advanced: Required. Add approvals and anti-reidentification rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `external_sharing_rules`, `notes`, `export_limits`
- If any export_allowed == true lacks redaction and audit_event → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RPT
- Pass conditions:
    - required_fields_present == true
    - matrix_count >= 20
    - default_deny_present == true
    - export_rules_consistent == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

RPT-06

RPT-06 — Reporting Data Quality Rules (reconciliation, correctness)

Header Block

- template_id: RPT-06
- title: Reporting Data Quality Rules (reconciliation, correctness)
- type: reporting_aggregations
- template_version: 1.0.0
- output_path: 10_app/reporting/RPT-06_Reporting_Data_Quality_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RPT
- upstream_dependencies: ["RPT-02", "RPT-03", "DQV-02"]
- inputs_required: ["RPT-02", "RPT-03", "DQV-02", "DQV-06", "BI-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how reporting correctness is verified: reconciliation rules between raw sources and aggregates/snapshots, acceptable deltas, auditability of metric computations, and actions when discrepancies are detected.

Inputs Required

- RPT-02: {{xref:RPT-02}} | OPTIONAL
- RPT-03: {{xref:RPT-03}} | OPTIONAL
- DQV-02: {{xref:DQV-02}} | OPTIONAL

- DQV-06: {{xref:DQV-06}} | OPTIONAL
- BI-05: {{xref:BI-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, mark N/A.
- Reconciliation rules catalog (minimum 15 rules)
- For each rule:
    - recon_id
    - metric_id
    - source of truth dataset(s)
    - reporting dataset (rollup/snapshot)
    - reconciliation method (exact match, bounded delta)
    - acceptable delta (percent/absolute)
    - schedule (daily/weekly)
    - owner
    - failure action (alert, block release, recompute)
    - evidence artifact (report)
- Global rules:
    - how to handle late data affecting reports
    - recompute policy linkage (RPT-04)
- Verification checklist

Optional Fields

- Audit-ready evidence pack outline | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block only.
- Every metric used in reporting must have at least one reconciliation rule.
- Acceptable deltas must be explicit and justified.
- Failures must have deterministic actions and owners.

Output Format

1) Applicability

- applies: {{recon.applies}} (true/false)
- 00_NA (if not applies): {{recon.na_block}} | OPTIONAL

2) Reconciliation Rules (canonical)

recon_id	metric_id	source_truth	reporting_target	method	acceptable_delta	schedule	owner	action	evidence	notes
rec_01	{{rules[0].metric}}	{{rules[0].source}}	{{rules[0].target}}	{{rules[0].method}}	{{rules[0].delta}}	{{rules[0].schedule}}	{{rules[0].owner}}	{{rules[0].action}}	{{rules[0].evidence}}	{{rules[0].notes}}
rec_02	{{rules[1].metric}}	{{rules[1].source}}	{{rules[1].target}}	{{rules[1].method}}	{{rules[1].delta}}	{{rules[1].schedule}}	{{rules[1].owner}}	{{rules[1].action}}	{{rules[1].evidence}}	{{rules[1].notes}}

3) Global Rules (required if applies)

- Late data handling: {{global.late_data}}
- Recompute linkage pointer: {{xref:RPT-04}} | OPTIONAL
- Escalation path: {{global.escalation}} | OPTIONAL

4) Verification Checklist (required if applies)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DQV-02}} | OPTIONAL, {{xref:RPT-04}} | OPTIONAL
- Downstream: {{xref:ALRT-*}} | OPTIONAL, {{xref:RELOPS-05}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL